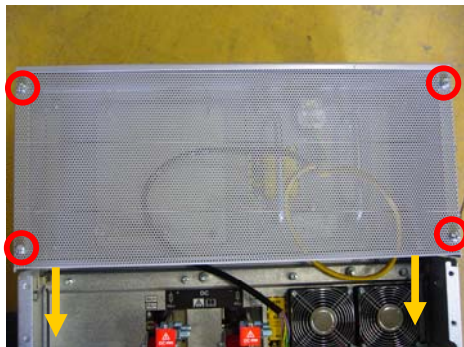


Transformer Front Cover



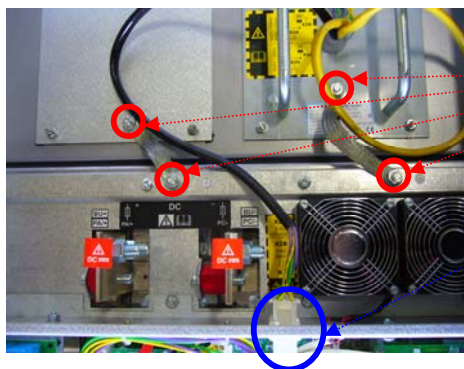
Remove 4 screws.

The front cover can be removed.



Size	Torque
M6x12	5.5Nm

Dismantling / reassembling Transformer connection



Remove 4 nuts.

Disconnect 1 wire:
X2B ->X2

The Transformer can be removed.



Size	Torque
M8	13.5Nm

Plastic Parts KIT: VY1A1410



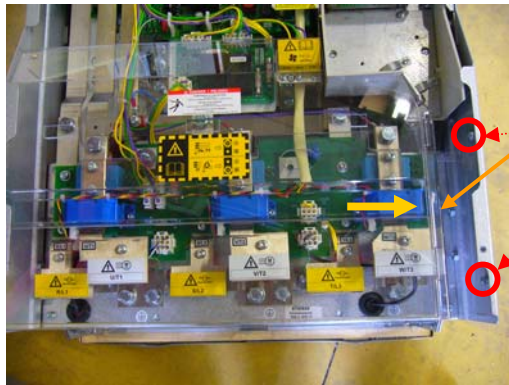
Remove 3 screws

The Plastic Parts kit can be removed.



Size	Torque
M4x12	1.2Nm

Plastic Parts KIT: VY1A1410



Push here and disengage plastic part

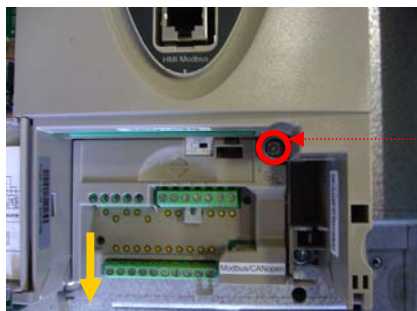
Remove 2 nuts

The Plastic parts KIT can be removed.



Size	Torque
M6	5.5Nm

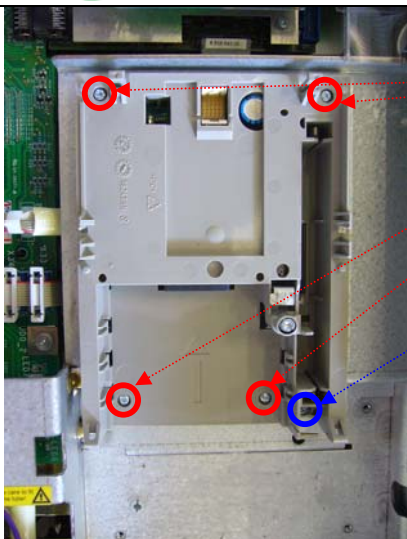
Control bloc: VX4A61101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.

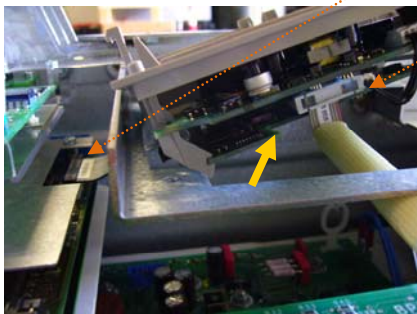


Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5-S8).

Remove 1 screw (S38).



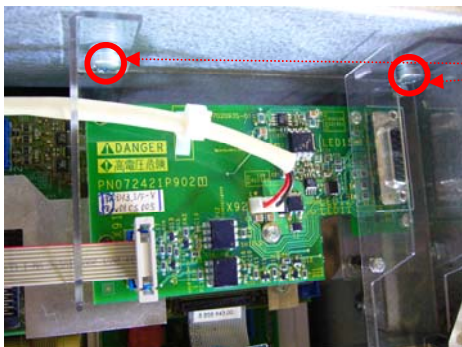
Disconnect the ribbon cable.
X3 from Control bloc "interface Board" -> X3 on Motor Control Board

Disconnect the ribbon cable.
X4 from Control bloc "interface Board" -> X4 on Measuring Board

The Control bloc can be removed.

Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5Nm

Braking Unit KIT: VZ3F1112



Remove 2 screws

Disconnect the wires. From right to left:
White X92 -> X92 on measuring board (W28)
Ribbon X91 -> X93 on measuring board (W27)

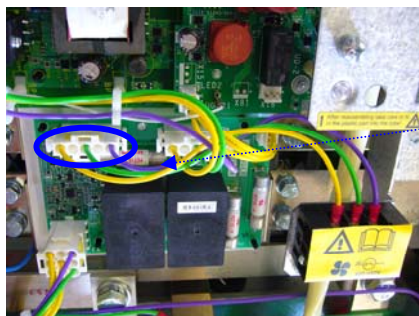
The Braking Unit KIT can be removed.

Size	Torque
M6x14	5.5Nm

Metal Support for Control bloc



Remove 2 nuts



Disconnect the wire:

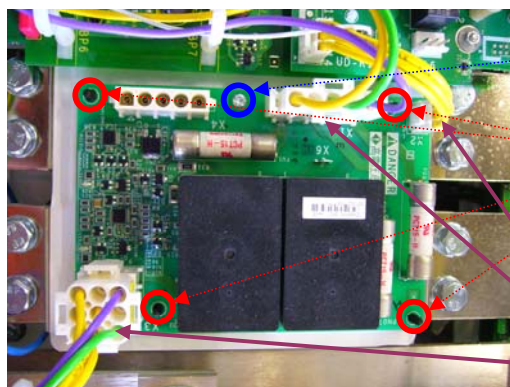
Yellow/Green/Purple TB1 -> X4 on fan control board

The metal support can be removed.



Size	Torque
M6	5.5Nm

Fan Control Board: VX5A1400



Remove 1 screw

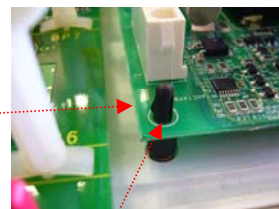
Push on the 4 plastic supports to remove the board.

Disconnect the wires. From right to left:

Yellow X2->X14 on measuring board

Yellow/Green/Purple X1-> X2 on metal kit ??->X2B on self-> X2 on RFI filter board

Yellow/Green/Purple X3-> turbine alimentation on RFI filter board



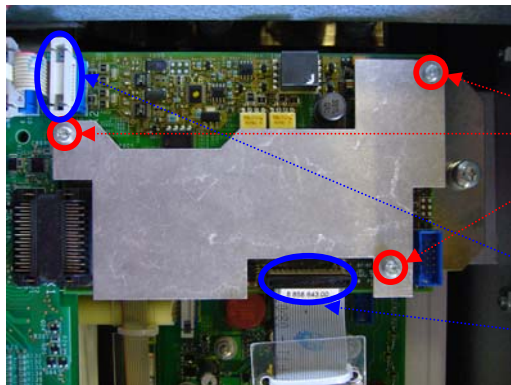
Push on the side.

The Fan Control Board can be removed.



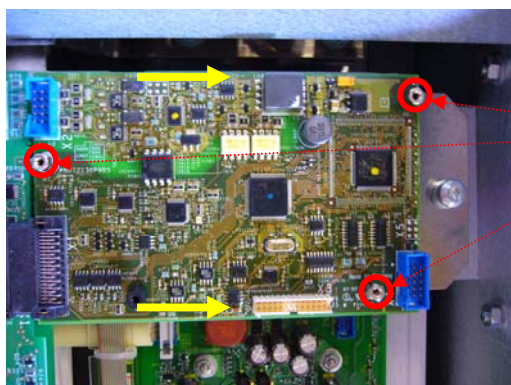
Size	Torque
M3x6	0.8Nm
M3x8	clips

Motor Control Board: VX4A61101Y



Remove the 3 screws. (P1)

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on measuring board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs (P2)

Be careful, at the time of reassembling;
don't forget to put the steel.



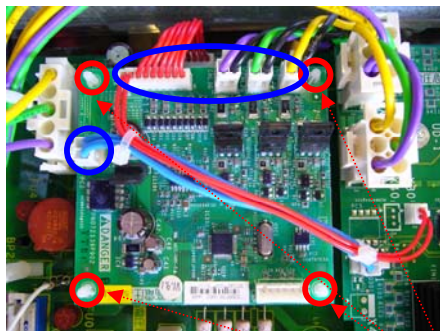
Remove the 2 screws. (P3)

The Motor Control Board can be removed.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm
P3	M3X6	0.8Nm

Soft Charge board: VX5A1301



Disconnect the 5 wires, *left to the right*:

Blue CN7A-> X31 on measuring board

Red CN2A-> X30 on measuring board and X30 on power board

Purple/Black CNL3G -> Gate rectifier 3 on I3

Green/Black CNL2G -> Gate rectifier 2 on L2

Yellow/Black CNL1G -> Gate rectifier 1 on L1



Push on the 4 plastic supports
to remove the board.

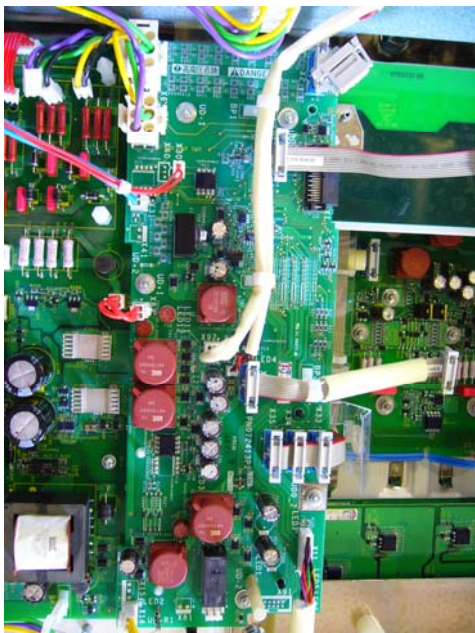
Push on the side.

The Soft charge Board can be removed.



Size	Torque
M3x8	clips

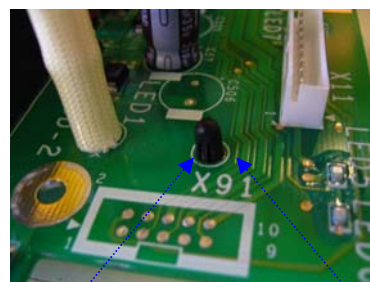
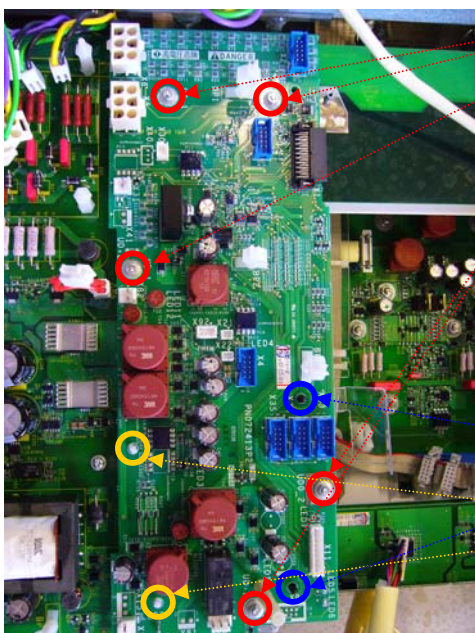
Measuring Board: VX5A71HC200250



Disconnect the 15 wires, *unclockwise*:

Ribbon X2-> X2 on motor control board
 Yellow/Green/Purple X6-> bus bar U Yellow, bus bar V Green, bus bar W Purple
 Yellow/Green/Purple X5->X6 on measuring board
 Red X30-> CN2A on Soft charge board
 Blue X31-> CN7A on soft charge board
 Red X82-> X82 on power Board
 Yellow X14 -> X2 on Fan control board
 White UD-R1 -> discharging resistor
 Multicolour X11-> currents sensors U V W
 Ribbon X35-> X33 on Gate drive board W
 Ribbon X34-> X33 on Gate drive board V
 Ribbon X33-> X33 on Gate drive board U
 Ribbon X4-> X4 on control block
 White X22-> inside Fan alimentation
 White X21-> inside Fan alimentation
 White X92-> X92 on braking unit device
 Ribbon X93-> X91 on braking unit device

Remove the 5 screws. (P1)



Push on the 2 plastic supports to remove the board. (P2) Push on the side.

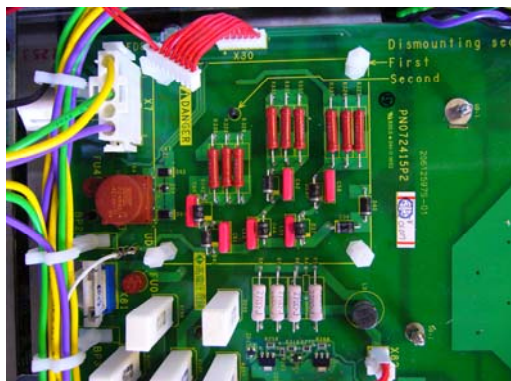
Push on the 2 plastic supports to remove the board (P3)

The Measuring board can be removed.



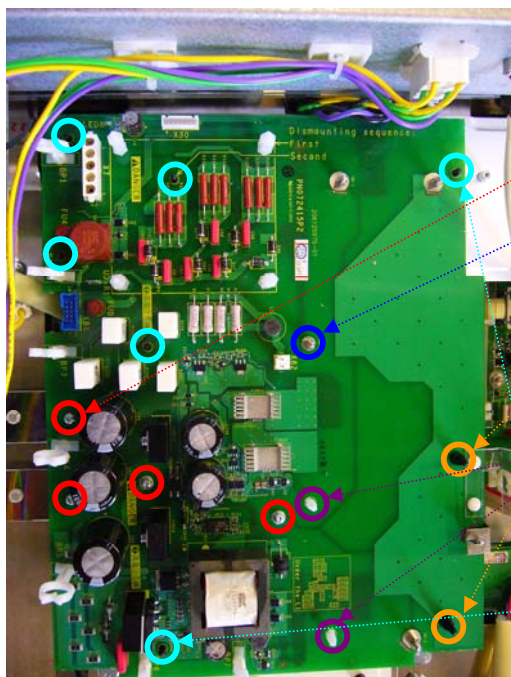
Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x8	clips
P3	M3x8	clips

Power Board: VX5A1HC2540



Disconnect the 4 wires, *unclockwise*:

Red X30 -> CN2A on softcharge board
Yellow/Green/Purple X7 -> X7 on rectifier snubber circuit
White UD+R1 -> discharging resistor
Ribbon X81 -> X82 on Gate drive IGBT U
Red X82 -> X82 on measuring board



Remove the 4 screws. (P1)

Remove the 1 screw. (P2)

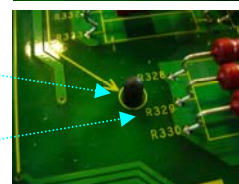
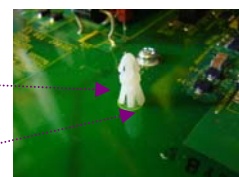
Remove the 2 screw. (P3)

Push on the 2 plastic supports to remove the board. (P4)

Push on the side.

Push on the 2 plastic supports to remove the board (P3)

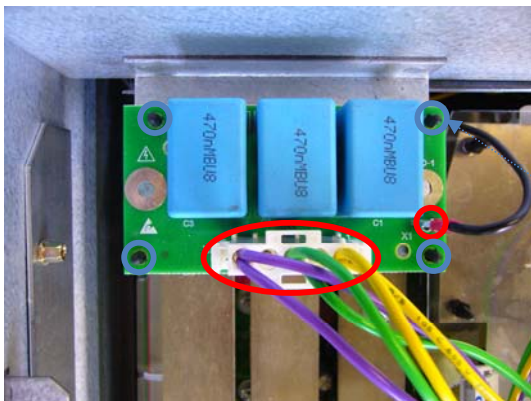
Push on the side.



The Power board can be removed.

Mark	Size	Torque
P1	M3x6	0.8Nm
P3	M3x8	Clips & 0.8Nm
P4	M3x8	clips

Rectifier Snubber Circuit: VX4A1206



Disconnect 2 Wires left to right

Yellow/Green/Purple X7->X7 on power board and X1 on RFI filter board

Black X1-> bus bar PC

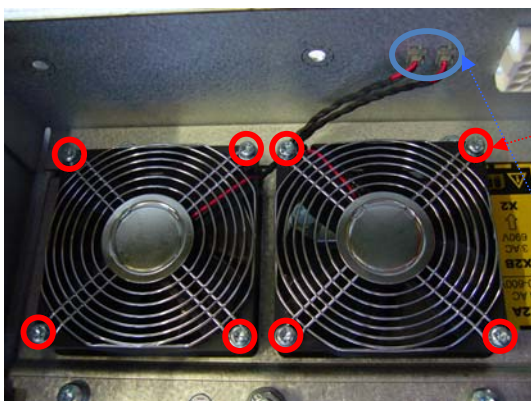
Push on the 4 plastic supports to remove the board.



Size	Torque
M3x8	clips

The Rectifier Snubber Circuit can be removed.

Internal Fan KIT: VZ3V1213



Remove 8 screws

Disconnect 2 Wires

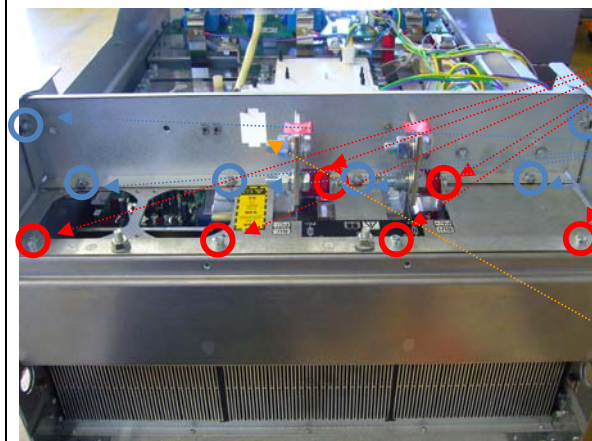
Black Fan-> X21 X22 on measuring board



Size	Torque
M4x35	1.2Nm

The Internal Fan KIT can be removed.

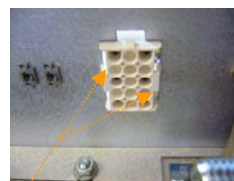
Metal Body Parts KIT



Remove 6 screws (P1).

Remove 6 screws (P2).

Push on the 2 plastic supports to remove the connector X2.

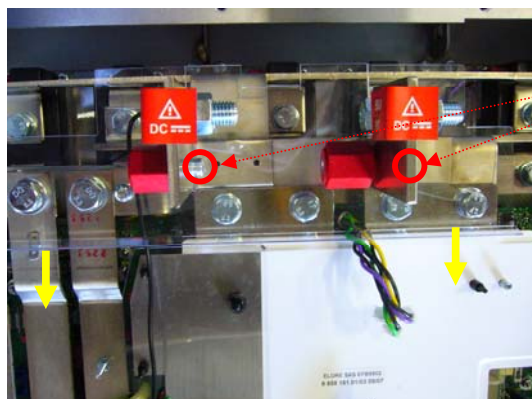


The Metal Body Parts KIT can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm

Plastic Parts KIT: VY1A1410



Remove 2 screws

Don't forget to replace it during reassembling

The Plastic parts KIT can be removed.



Size	Torque
M6X14	5.5Nm

DC bus bar kit: VZ3N1342



Remove 4 screws (P1)

Remove 1 screw (P2)

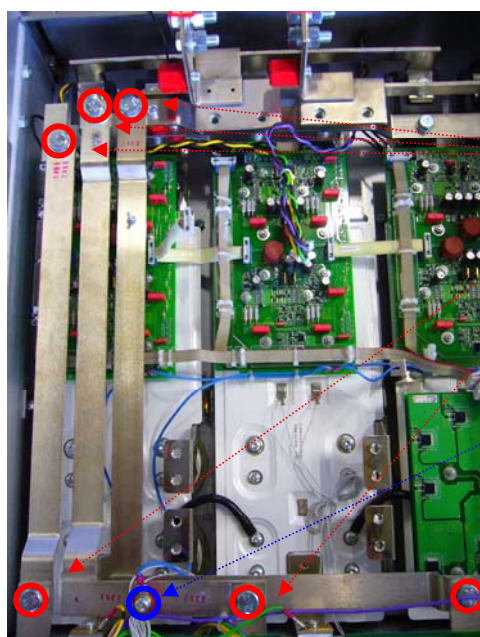
Remove 12 screws (P3)



Mark	Size	Torque
P1	M10X20	27Nm
P2	M6X12	5.5Nm
P3	M8x20	13.5Nm

The DC bus bar kit can be removed.

AC Bus Bar KIT: VZ3N1345



Remove 6 screws (J110, J120, J122).

Remove 1 screw (J125, J159, J22) and Disconnect 1 wire

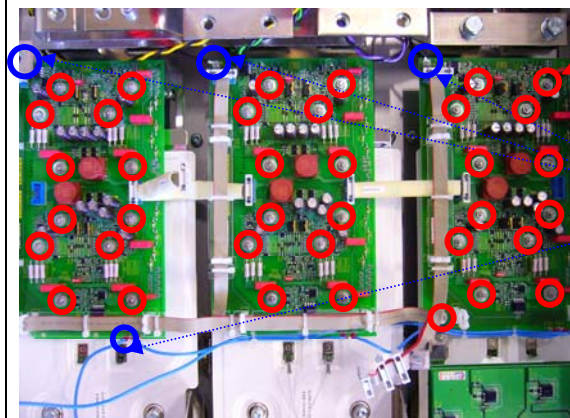
Blue -> X52 on Gate drive board IGBT W



Mark	Size	Torque
J110 J120 J122	M10x25	27Nm
J125 J159 J22	M6x12	5.5Nm

The AC Bus Bar KIT can be removed.

Gate Drive Board IGBT: VX5A1206



Remove 37 nuts

Disconnect the 4 wires top to down.

Black X4 on gate drive Board IGBT U/V/W-> thermal sensor

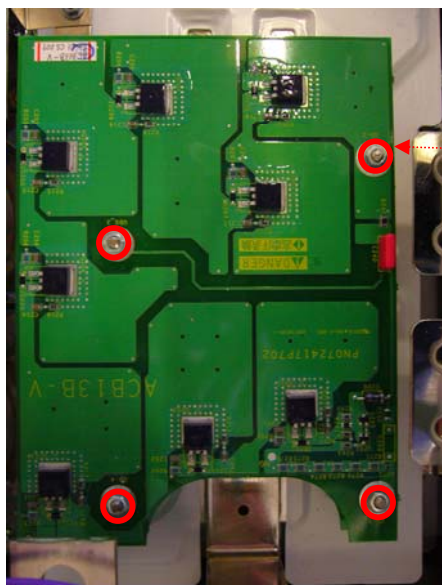
Blue X51 on gate drive Board IGBT U->Power Interconnection bar U

The Gate Drive Boards IGBT can be removed.



Size	Torque
M4	1.2Nm

Active Balancing Board: VX5A1105



Remove 4 nuts

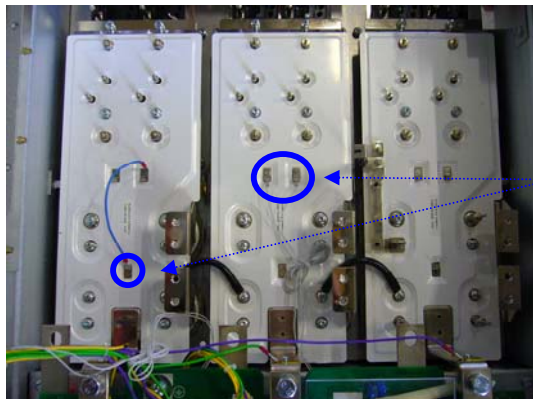
Be careful, at the time of reassembling;
don't forget to put both cards.

The Active Balancing Board can be removed.



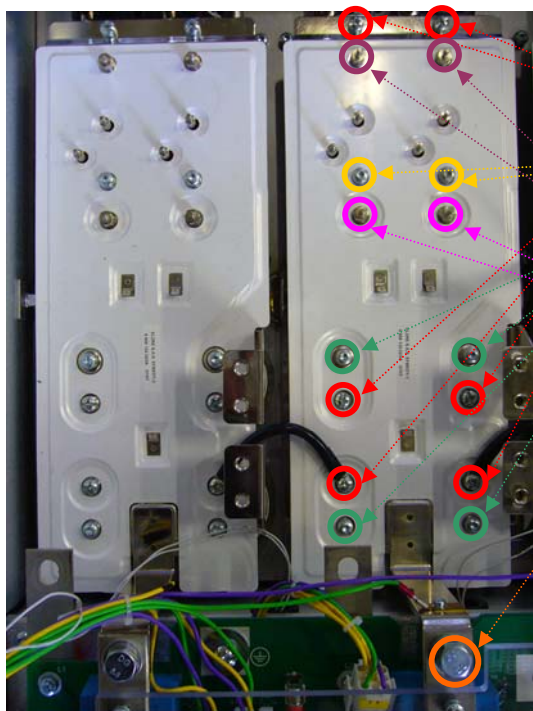
Size	Torque
M4	1.2Nm

DC bus bar kit: VZ3N1342



Disconnect the 4 wires Left to right.

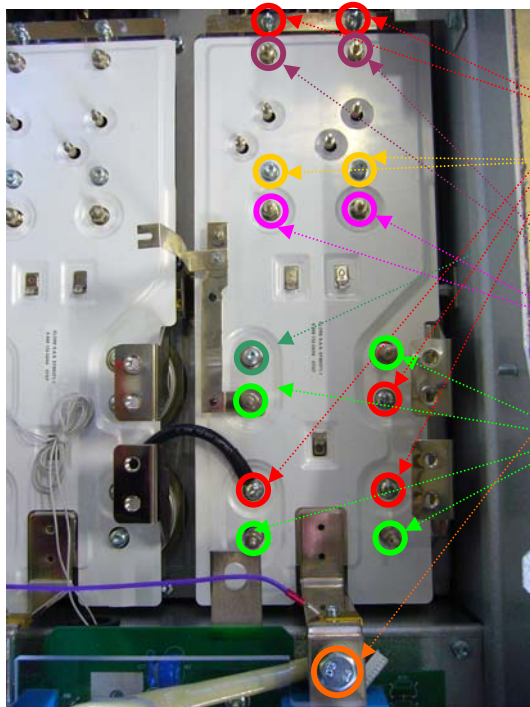
Blue wire on interconnection bar U -> X51 on Gate drive Board IGBT U
White wires on interconnection bar V -> discharging resistor



For 2 AC bus bar U V, you should remove:

- Remove 6 screws (J125).
- Remove 2 screws (J135)
- Remove 4 screws (J136)
- Remove 2 screws (J26)
- Remove 2 screws (J27)
- Remove 1 screw (J110, J120, J122)

Be careful, at the time of reassembling;
don't forget to put the 2 Shunt.



For this AC bus bar W, you should remove:

Remove 5 screws (J125).

Remove 2 screws (J135)

Remove 1 screw (J136)

Remove 2 screws (J26)

Remove 2 screws (J27)

Remove 1 screw (J110)

Remove 4 screws (J24)

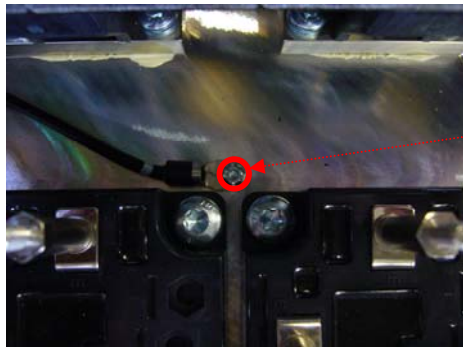
Be careful, at the time of reassembling;
don't forget to put the 2 Shunt.



Mark	Size	Torque
J125	M6x12	3,3Nm
J26	ST 10x6	3,3Nm
J135	M6x20	3,3Nm
J27	ST 10x11	3,3Nm
J136	M6x14	3.3Nm
J24	ST 10x25	3.3Nm
J110	M12x25	45Nm

The DC Bus Bar KIT can be removed.

Thermal Sensor: VZ3G1105



For each 3 branch U/V/W, you should remove:

Remove 1 screw

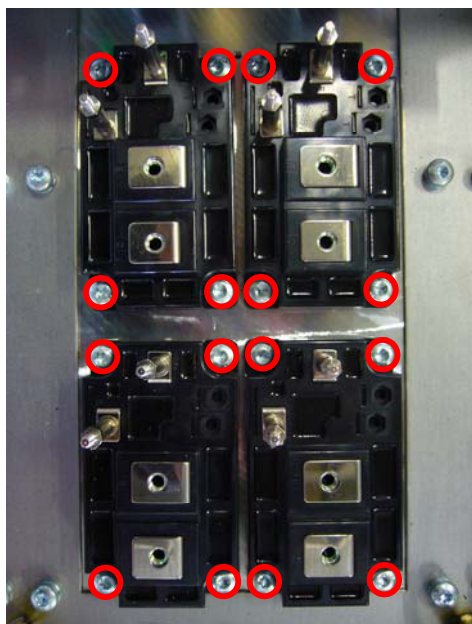
Be careful, don't forget to connect the wire on the connector "X4" of the Gate drive board IGBT



Size	Torque
M3x6	0.8Nm

The thermal sensor can be removed.

Lot of IGBT: VZ3IM1502M1771



For each 3 branch U/V/W, you should remove:

Remove 16 screws

It is **not necessary** to change all IGBT but just this one that is damaged.

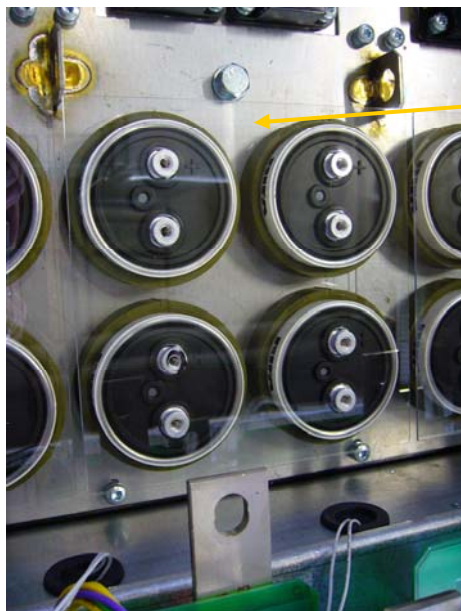
After changing, **be careful** to set up direction.
Don't forget applying the grease.



Size	Torque
M6x20	3.0Nm

The IGBT can be removed.

Lot of Capacitors: VY1ADC1117



First step

Remove plastic protection part for each 3 branch U/V/W



Second step

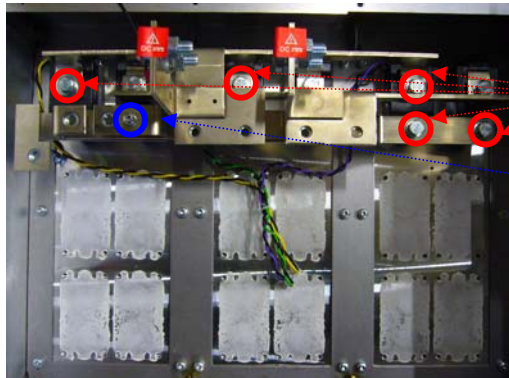
For each 3 branch U/V/W, you should remove 4 capacitors:

To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction

The Capacitors can be removed.

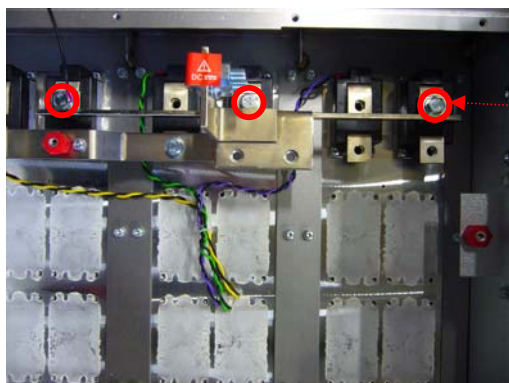
Rectifier module thyristor: VZ3TM2200M4371
Rectifier module diode: VZ3DM2200M5471
AC Bus BAR KIT : VZ3N1345



Part 1

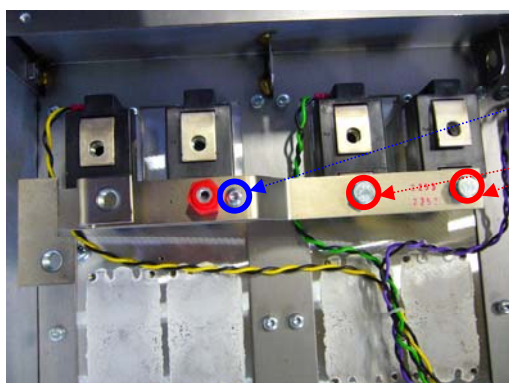
Remove 5 screws (J143)

Remove 1 screw (P1)



Part 2

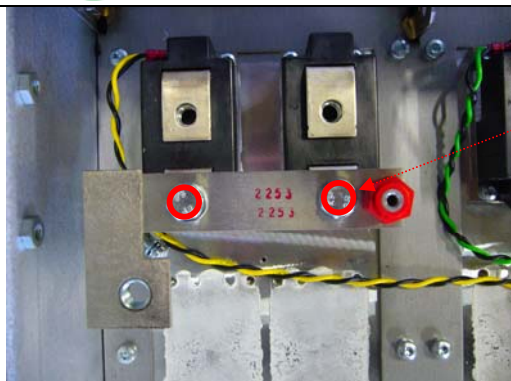
Remove 3 screws (J143)



Part 3

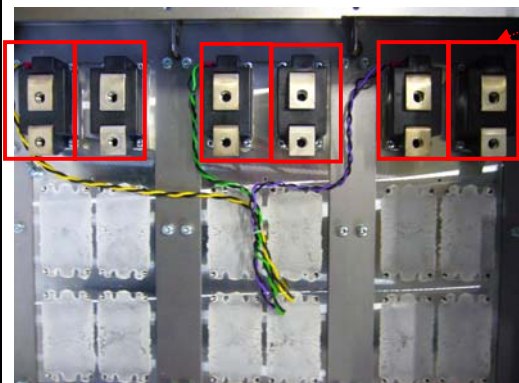
Remove 1 screw (P1)

Remove 2 screws (J143)



Part 4

Remove 2 screws (J143)



Remove the 24 screws (P2).

After changing, **be careful** to set up direction.

Don't forget applying the grease.



The Thyristor can be removed.

Mark	Size	Torque
J143	M10x25	13,5Nm
P1	M6x12	5.5Nm
P2	M5x25	5.0Nm

Rectifier module thyristor: VZ3TM2200M4371



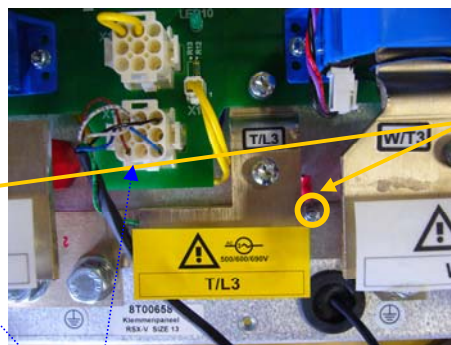
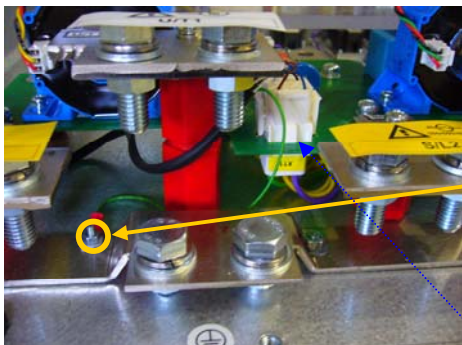
Be careful, after changing respect the colour line for the wires.

From left to right

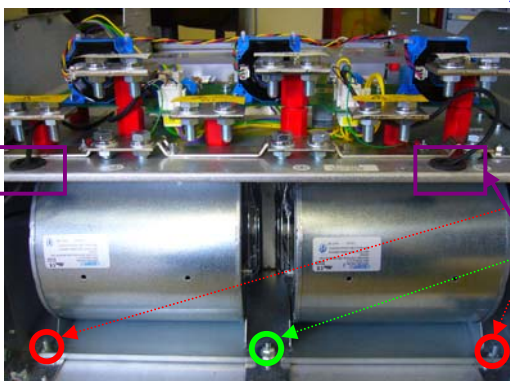
[Yellow-Black]- [Green-Black] - [Purple-Black]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Fan Power Electronic: VZ3V1212



Be careful, don't forget to disconnect the ground wire.



Disconnect 2 connectors
Fan turbine 1->X11 on RFI filter board->X3 on fan control board
Fan turbine 2->X13 on RFI filter board->X3 on fan control board

Remove 2 screws (J130)

Remove 1 screw (J139)

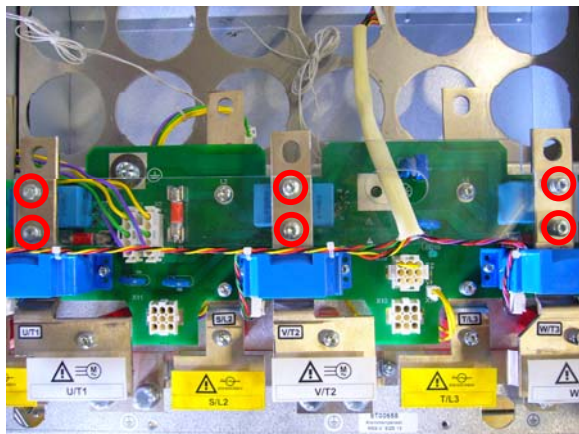
Be careful, to remove the fan, remove the gasket and slide the wire in the hole.



Mark	Size	Torque
J130	M6	5.5Nm
J139	M8	13.5Nm

The Fan Power Electronic can be removed.

Plastic Parts kit: VY1A1410



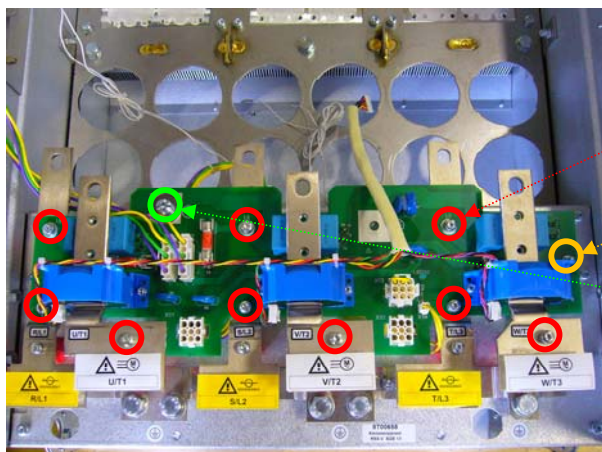
Remove 6 screws

The Plastic Parts KIT can be removed.



Size	Torque
M6x20	5.5Nm

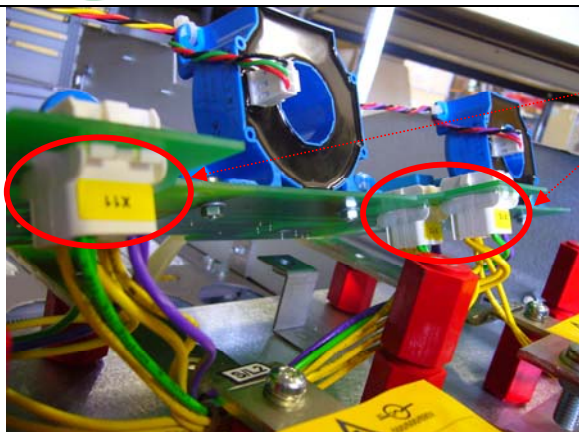
RFI Filter Board: VX4A1120



Remove 9 screws (P1)

Remove 1 screw (P2)

Remove 1 screw (P3)



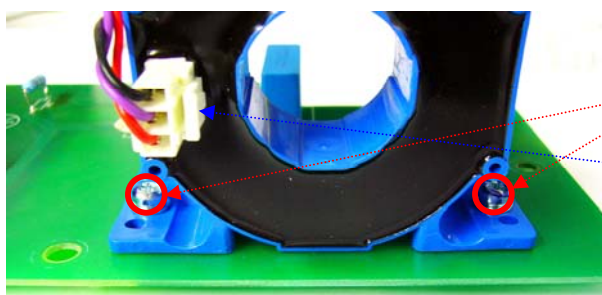
To change the board, don't forget to disconnect the 3 connectors X11 X12 and X13->Fan turbines



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm
P3	M6x12	5.5Nm

The RFI filter board can be removed.

Motor current sensor: VY1A1108



For each 3 branch U/V/W, you should remove:

Remove 2 screws

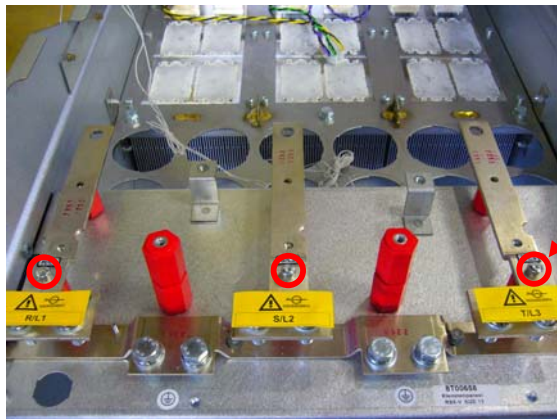
Disconnect 1 wire
Current sensor->X11 on measuring board



Size	Torque
M4x12	1.2Nm
M4	1.2Nm

The Motor current sensor can be removed.

AC Bus BAR KIT : VZ3N1345



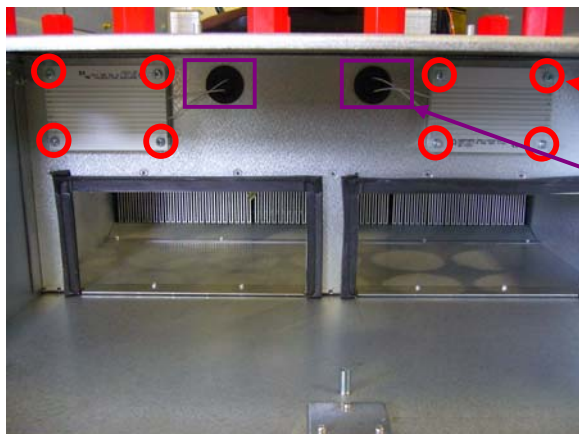
Remove 3 screws

The AC Bus Bar KIT can be removed.



Size	Torque
M6x12	5.5Nm

Discharging Resistor VZ3R36KW125



Remove 8 screws (J123)

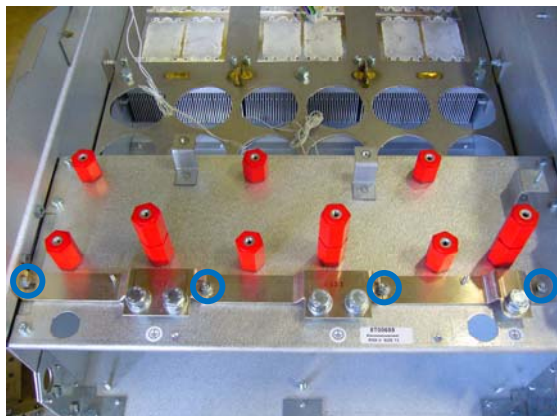
Be careful, to remove the discharging resistor, remove the gasket and slide the wire in the hole.

The discharging resistor can be removed.



Mark	Size	Torque
J123	M4x20	1.2Nm

AC Bus BAR KIT: VZ3N1345



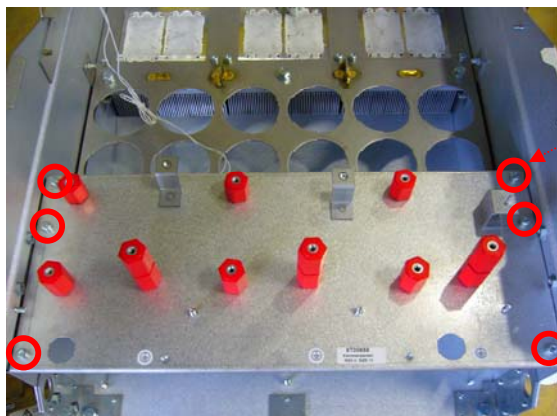
Remove 4 nuts

The Ground bar can be removed.



Size	Torque
M6	5.5Nm

Metal Body Part KIT



Remove 6 screws

The Metal body parts KIT can be removed.



Size	Torque
M6x14	5.5Nm



9.29.2 Product Assembling Drawing

No information

9.29.3 Product Cabling Drawing

Refer to following file: [Cabling diagram size 13.pdf part1](#)

Refer to following file: [Cabling diagram size 13.pdf part2](#)

9.30 ATV61/71 Size 15V (size, refer to 1.2)

9.30.1 Dismantling and reassembling

Size 15V: ATV71HC40Y, ATV71HC50Y, ATV71HC63Y, ATV61HC50Y, ATV61HC63Y, ATV61HC80Y

ATV71HC40Y	
Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1349	Wires KIT
VZ3N1346	AC Bus Bar KIT
VZ3N1343	DC Bus Bar KIT
VZ3IM1504M1771	Lot of 4 IGBT modules related (500A / 1700V
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1114	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100μF)
VY1A1411	Plastic Parts KIT
VY1A1309	Assembling KIT
VX5A71HC400500	Measuring Board
VX5A1HC6380	Power board
VX5A1301	Soft Charge Board
VX5A1207	Gate Drive Board
VX5A1106	Active Balancing Board
VX4A71101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1121	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VY1A1109	Motor Current Sensor
VY1A1217	Front Cover with I/O Terminal Cover
VZ3F1113	Braking Unit KIT
VZ3V1213	Internal Fan
VX4A1103	Front cover 4x7 Digits

ATV71HC50Y	
Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1349	Wires KIT
VZ3N1346	AC Bus Bar KIT
VZ3N1343	DC Bus Bar KIT
VZ3IM1504M1771	Lot of 4 IGBT modules related (500A / 1700V
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1114	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100μF)
VY1A1411	Plastic Parts KIT
VY1A1309	Assembling KIT
VX5A71HC500630	Measuring Board
VX5A1HC6380	Power board
VX5A1301	Soft Charge Board
VX5A1207	Gate Drive Board
VX5A1106	Active Balancing Board
VX4A71101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1121	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VZ3V1213	Internal Fan
VX5A1400	Fan Control Board
VZ3F1113	Braking Unit KIT
VX4A1103	Front cover 4x7 Digits

ATV71HC63Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1349	Wires KIT
VZ3N1346	AC Bus Bar KIT
VZ3N1343	DC Bus Bar KIT
VZ3IM1504M1771	Lot of 4 IGBT modules related (500A / 1700V
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1114	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1411	Plastic Parts KIT
VY1A1309	Assembling KIT
VX5A71HC630800	Measuring Board
VX5A1HC6380	Power board
VX5A1301	Soft Charge Board
VX5A1207	Gate Drive Board
VX5A1106	Active Balancing Board
VX4A71101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1121	RFI Filter Board
VZ3V1212	Fan Power Electronic
VZ3V1213	Internal Fan
VX5A1400	Fan Control Board
VZ3F1113	Braking Unit KIT

ATV61HC50Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1349	Wires KIT
VZ3N1346	AC Bus Bar KIT
VZ3N1343	DC Bus Bar KIT
VZ3IM1504M1771	Lot of 4 IGBT modules related (500A / 1700V
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1114	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1411	Plastic Parts KIT
VY1A1309	Assembling KIT
VX5A71HC400500	Measuring Board
VX5A1HC6380	Power board
VX5A1301	Soft Charge Board
VX5A1207	Gate Drive Board
VX5A1106	Active Balancing Board
VX4A61101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1121	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1103	Front cover 4x7 Digits
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VY1A1109	Motor Current Sensor
VY1A1217	Front Cover with I/O Terminal Cover
VZ3F1113	Braking Unit KIT
VZ3V1213	Internal Fan

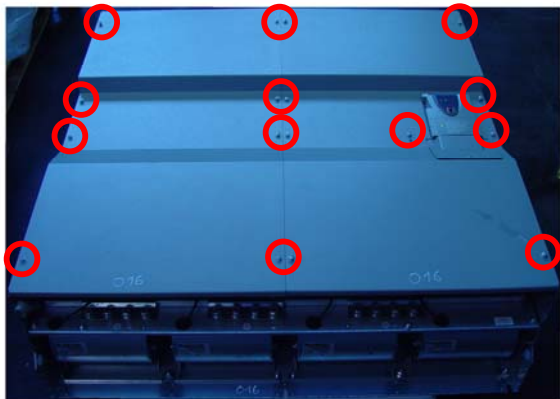
ATV61HC63Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1349	Wires KIT
VZ3N1346	AC Bus Bar KIT
VZ3N1343	DC Bus Bar KIT
VZ3IM1504M1771	Lot of 4 IGBT modules related (500A / 1700V
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1114	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1411	Plastic Parts KIT
VY1A1309	Assembling KIT
VX5A71HC500630	Measuring Board
VX5A1HC6380	Power board
VX5A1301	Soft Charge Board
VX5A1207	Gate Drive Board
VX5A1106	Active Balancing Board
VX4A61101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1121	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VY1A1109	Motor Current Sensor
VY1A1217	Front Cover with I/O Terminal Cover
VZ3F1113	Braking Unit KIT
VZ3V1213	Internal Fan
VX4A1103	Front cover 4x7 Digits

ATV61HC80Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1349	Wires KIT
VZ3N1346	AC Bus Bar KIT
VZ3N1343	DC Bus Bar KIT
VZ3IM1504M1771	Lot of 4 IGBT modules related (500A / 1700V
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1114	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1411	Plastic Parts KIT
VY1A1309	Assembling KIT
VX5A71HC630800	Measuring Board
VX5A1HC6380	Power board
VX5A1301	Soft Charge Board
VX5A1207	Gate Drive Board
VX5A1106	Active Balancing Board
VX4A61101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1121	RFI Filter Board
VX4A1104	Terminal Board
VZ3V1212	Fan Power Electronic
VX5A1400	Fan Control Board
VY1A1109	Motor Current Sensor
VY1A1217	Front Cover with I/O Terminal Cover
VZ3F1113	Braking Unit KIT
VZ3V1213	Internal Fan
VX4A1103	Front cover 4x7 Digits

Front Cover with I/O Terminal Cover: VY1A1217



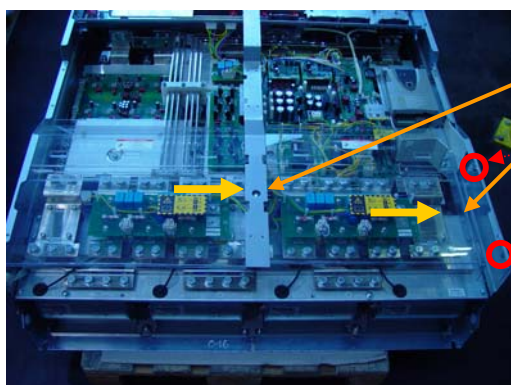
Remove 17 screws

The front cover can be removed.



Size	Torque
M6x12	5.5Nm

Plastic Parts KIT: VY1A1411



Push here and disengage plastic part

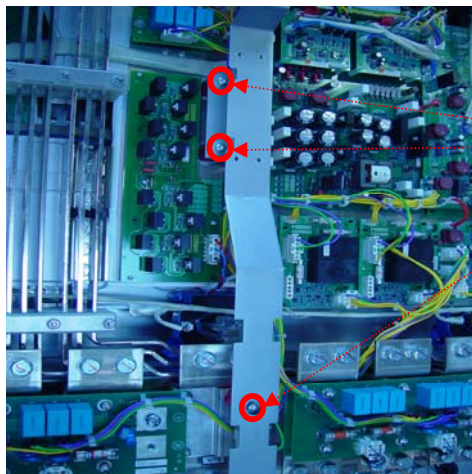
Remove 2 nuts

The Plastic part can be removed.

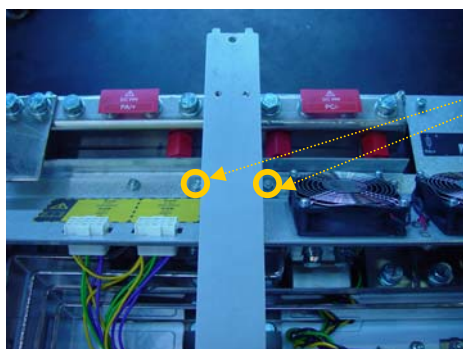


Size	Torque
M6	5.5Nm

metal kit.



Remove 3 screws (P1)



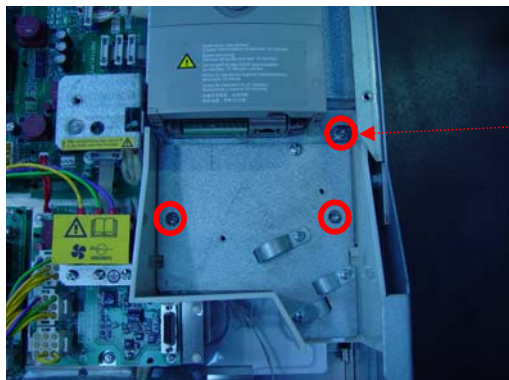
Remove 2 nuts (P2)



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm

The metal part can be removed.

Plastic Parts KIT: VY1A1411



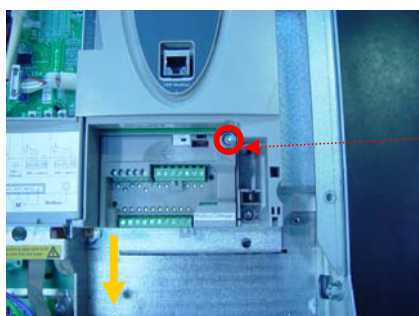
Remove 3 screws.

The kit can be removed.



Size	Torque
M4x12	1.2Nm

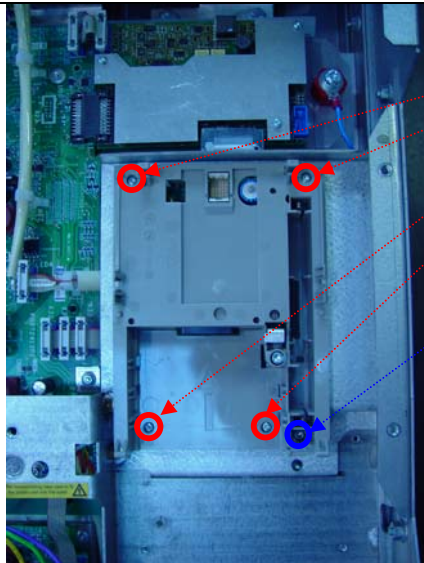
Control bloc: VX4A71101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.



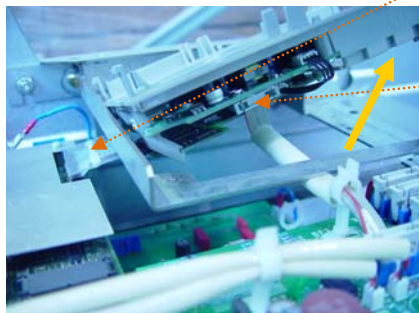
Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 to S8).

Remove 1 screw (S38).

Disconnect the ribbon cable.
X3 from Control bloc "interface Board" -> X3 on Motor Control Board



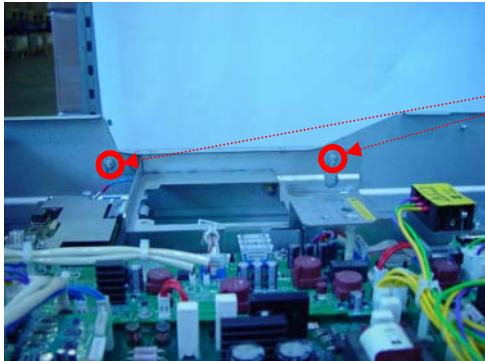
Disconnect the ribbon cable.
X4 from Control bloc "interface Board" -> X4 on Measuring Board

The Control bloc can be removed.

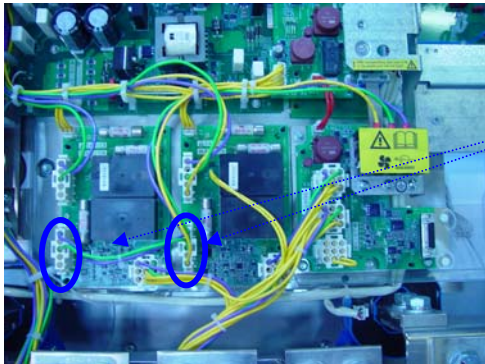


Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5NM

Metal Support for Control bloc



Remove 2 nuts



Disconnect wire:

Yellow/Green/Purple TB1 -> X4 on both fan control board

The metal support can be removed.



Size	Torque
M6	5.5Nm

Braking Unit KIT: VZ3F1113



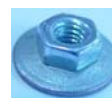
Disconnect 5 wires top to down:

- Red X8 -> X81 on measuring board
- Ribbon X91 -> X91 on measuring board
- Yellow/Green/Purple X3 -> X3 on fan control board 2
- Yellow/Green/Purple X33 -> X11 & X12 on RFI filter board 1
- Yellow X2 -> Shunt X2 on braking unit device



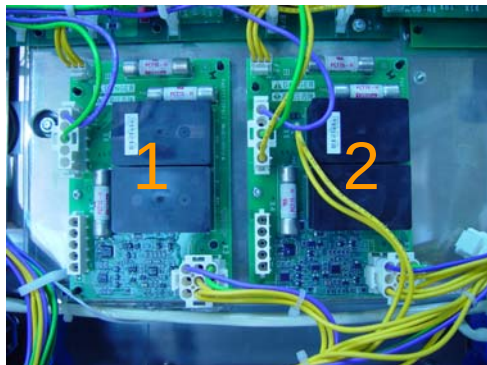
Remove 4 nuts

The Braking Unit Device can be removed.



Size	Torque
M3	0.8Nm

Fan Control Board. VX5A1400



Disconnect wires. From top to down:

Fan control Board 1

Yellow X2->X15 on measuring board

Yellow/Green/Purple X1->X2.1 on metal **kit ?**

Yellow/Green/Purple X3-> X13 X14 on RFI filter board 2

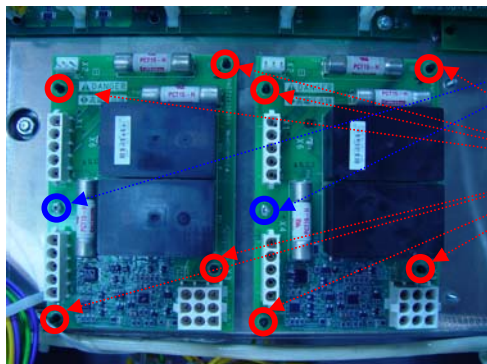
Fan control Board 2

Yellow X2->X14 on measuring board

Yellow/Green/Purple X1->X2.2 on metal **kit ?**

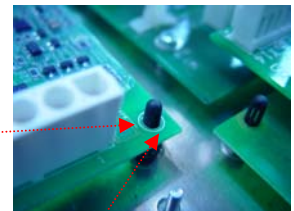
Yellow X6-> X11 & X14 on both RFI filter board

Yellow/Green/Purple X3-> X3 on braking unit device



Remove 2 screws

Push on the 8 plastic supports to remove the board.



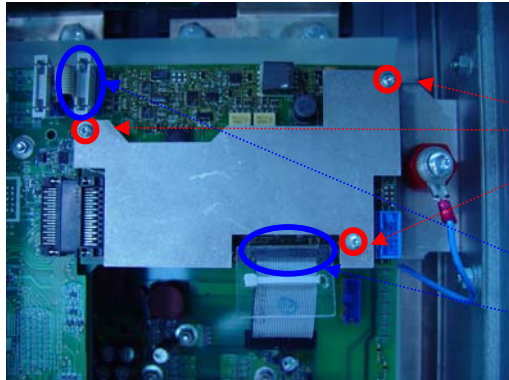
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

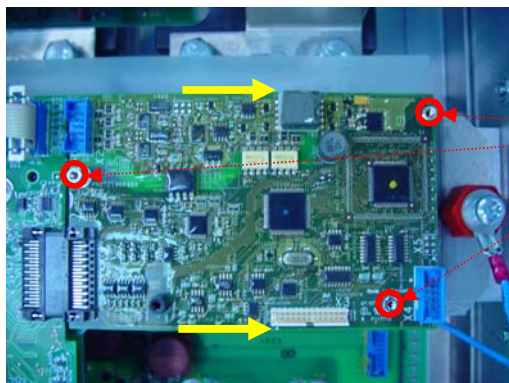
The Fan control board can be removed.

Motor Control Board: VX4A61101Y



Remove the 3 screws (P1).

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on measuring board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs (P2).

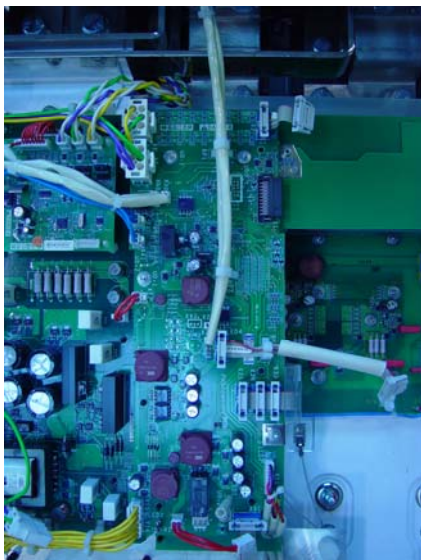
Be careful, at the time of reassembling;
don't forget to put the steel.

The Motor Control Board can be removed.



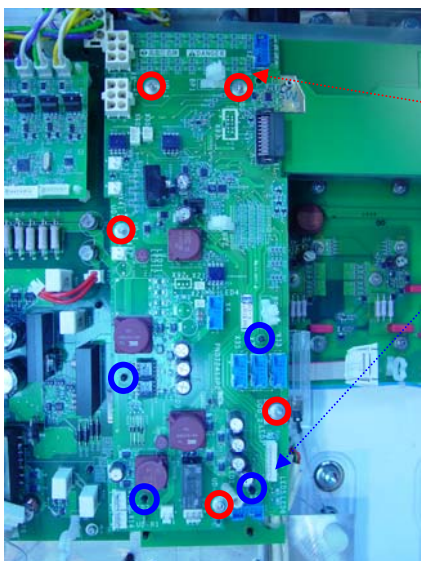
Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

Measuring board: VX5A71HC630800



Disconnect the 15 wires, *unlockwise*:

Ribbon X2-> X2 on motor control board
 Yellow/Green/Purple X6->X5 on measuring board
 Yellow/Green/Purple X5-> bus bar U Yellow, bus bar V Green, bus bar W Purple
 Red X30-> CN2A on Soft charge board 1
 Red X40-> CN2A on soft charge board 2
 Blue X31-> CN7A on soft charge board 1
 Blue X41-> CN7A on soft charge board 2
 Red X82-> X82 on power Board
 Yellow X15-> X2 on Fan control Board 1
 Yellow X14 -> X2 on Fan control board 2
 White UD-R1 -> discharging resistor
 Red X81-> X8 on Braking unit device
 Ribbon X91-> X91 on Braking unit device
 Multicolour X11-> currents sensors U V W
 Ribbon X35-> X33 on Gate drive board W
 Ribbon X34-> X33 on Gate drive board V
 Ribbon X33-> X33 on Gate drive board U
 Ribbon X4-> X4 on control block
 White X22-> inside Fan alimentation
 White X21-> inside Fan alimentation



Remove 5 nuts

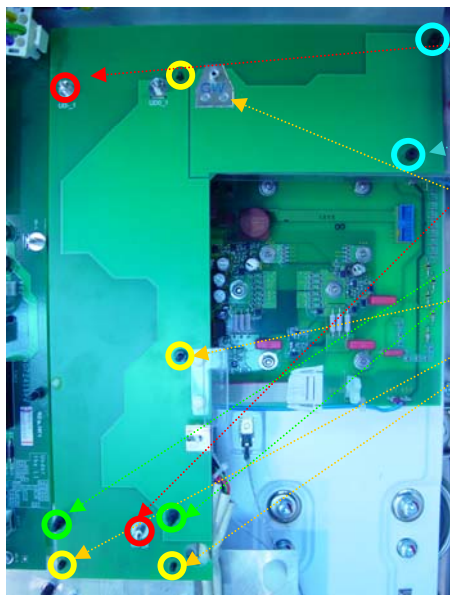
Push on the 4 plastic supports to remove the board.



Size	Torque
M3x8	clips
M4	1.2Nm

The measuring board can be removed.

Assembly KIT: VY1A1309



Remove 2 screws (P1)

Remove 2 screws (P2)

Remove 2 screws (P3)

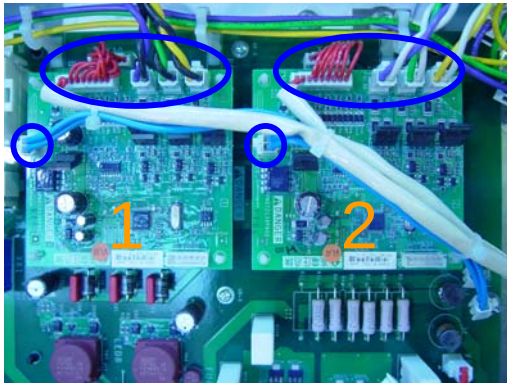
Push on the 4 plastic supports to remove the board. (P3)

The shielding plate PCB Measuring Board can be removed.



Mark	Size	Torque
P1	M3x5.5	0.8Nm
P2	M3x6	0.8Nm
P3	M3x8	clips

Soft Charge board: VX5A1301



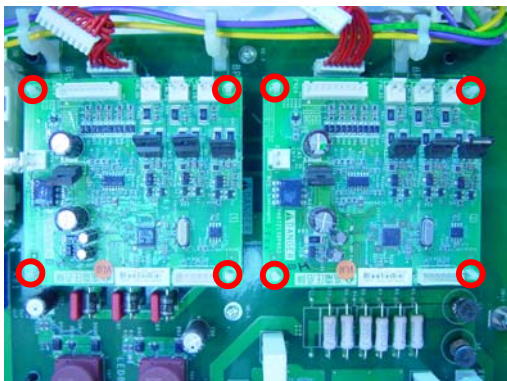
Disconnect wires, *left to the right*:

Soft charge Board 1

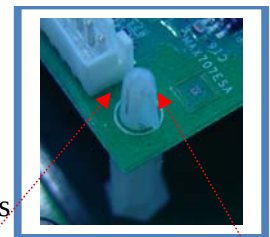
Blue CN7A-> X31 on measuring board
Red CN2A-> X30 on measuring board and X30 on power board
Purple/Black CNL3G ->Gate rectifier 3.1 on L3.1
Green/Black CNL2G ->Gate rectifier 2.1 on L2.1
Yellow/Black CNL1G -> Gate rectifier 1.1 on L1.1

Soft charge Board 2

Blue CN7A-> X41 on measuring board
Red CN2A-> X40 on measuring board and X40 on power board
Purple/White CNL3G ->Gate rectifier 3.2 on L3.2
Green/ White CNL2G ->Gate rectifier 2.2 on L2.2
Yellow/ White CNL1G -> Gate rectifier 1.2 on L1.2



Push on the 8 plastic supports
to remove the board.



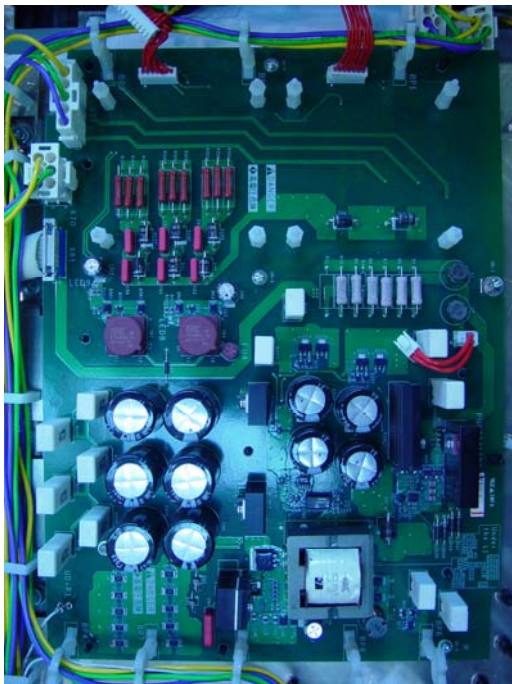
Push on the side.

The Soft charge Board can be removed.



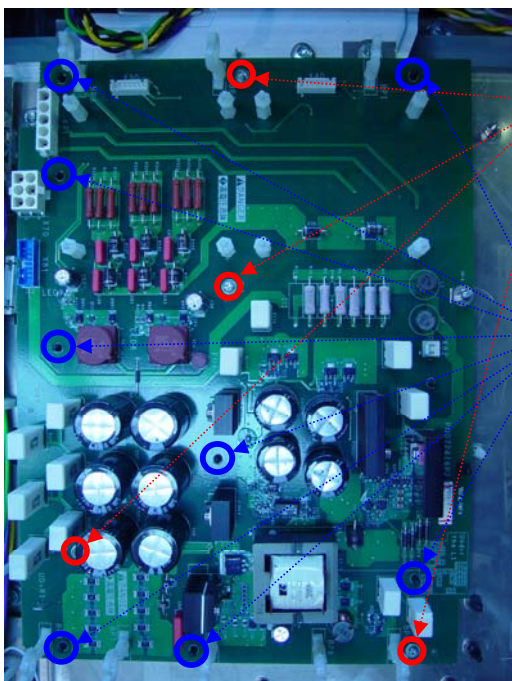
Size	Torque
M3x8	clips

Power Board: VX5A1HC6380



Disconnect wires, *unclockwise*:

- Red X40 -> CN2A on soft charge board 2
- Red X30 -> CN2A on soft charge board 1
- Yellow/Green/Purple X71 -> Yellow X1/Green X2/Purple X3 on Filter board 2
- Yellow/Green/Purple X70 -> X7 on rectifier snubber circuit
- Ribbon X81 -> X82 on Gate drive board IGBT U
- White UD+R1 -> discharging resistor
- Red X82 -> X82 on measuring board



Remove the 4 screws.

Push on the 8 plastic supports to remove the board



Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

The Power board can be removed.

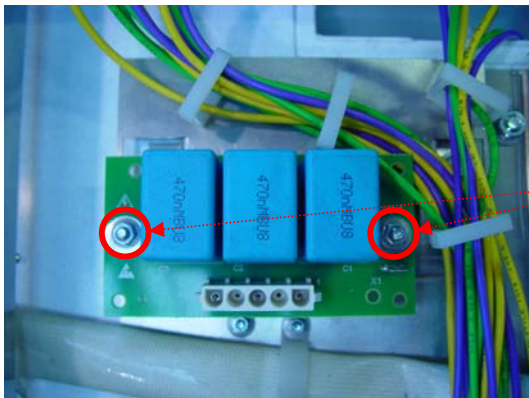
Rectifier Snubber Circuit: VX4A1206



Disconnect one wire

Yellow/Green/Purple X7->X70 on power board

->Yellow X1/Green X2/Purple X3 on Filter board 1



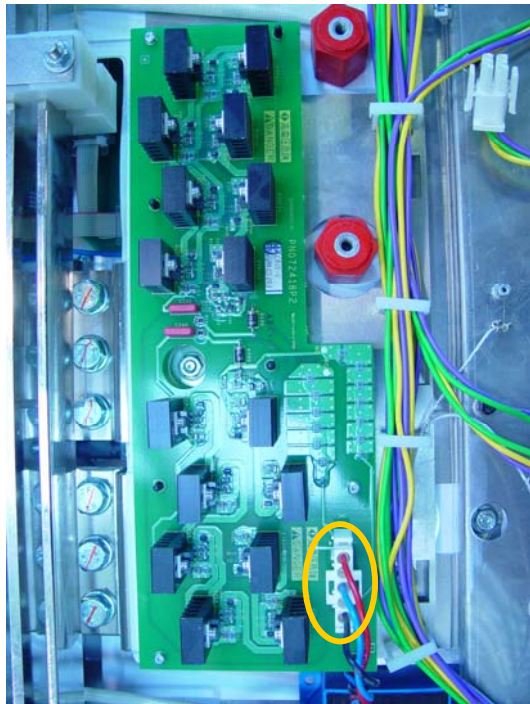
Remove 2 nuts

The Rectifier Snubber Circuit can be removed.



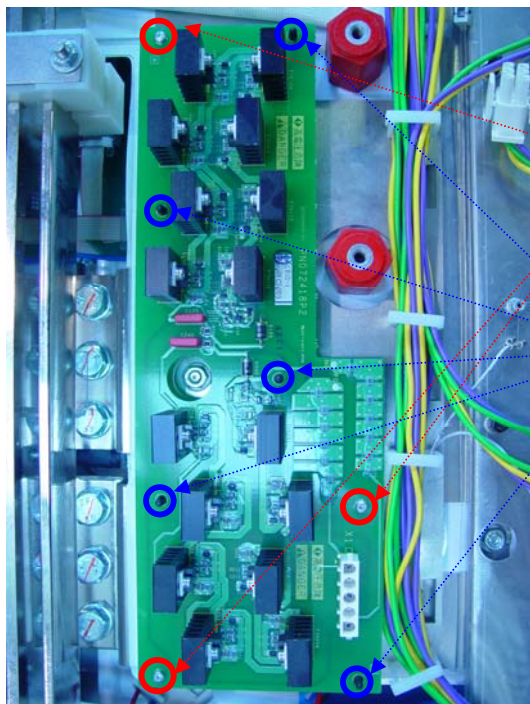
Size	Torque
M4	1.2Nm

Active current balancing board: VX5A1106



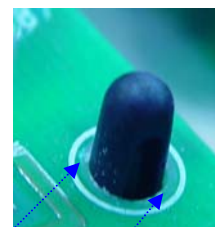
Disconnect one wire

Red/Blue/Black X1-> Bus bar V



Remove 3 screws

Push on the 5 plastic supports
to remove the board



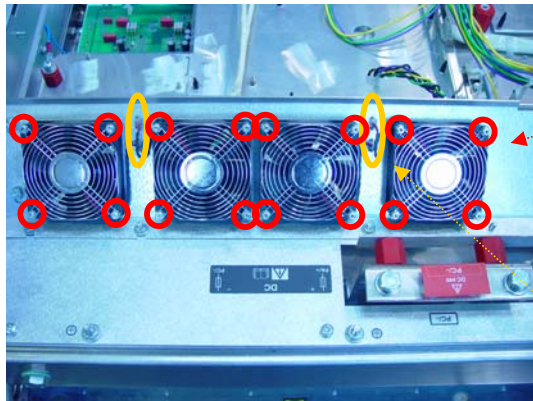
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

The active current balancing board can be removed.

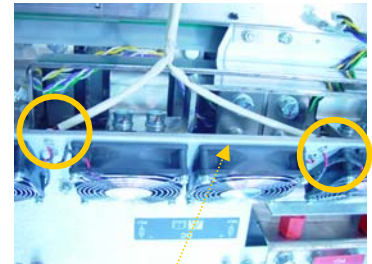
Internal fan kit: VZ3V1213



Remove 16 screws

Disconnect 4 Wires

Black Fan-> X21 X22 on measuring board



The Internal Fan can be removed.



Size	Torque
M4x35	1.2Nm

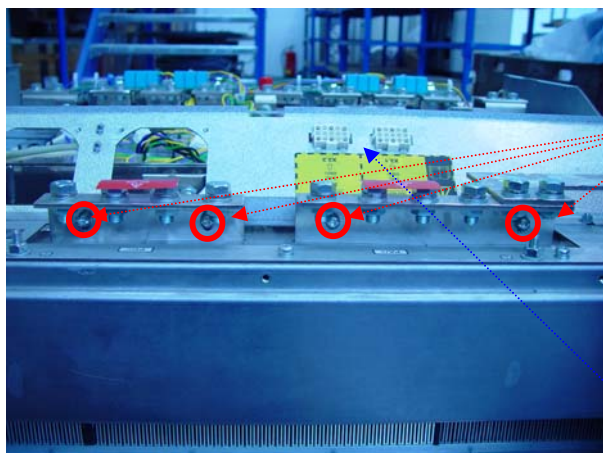
Metal Body Parts KIT



Step 1

Remove 2 screws (P3).

Remove 1 screw (P2)

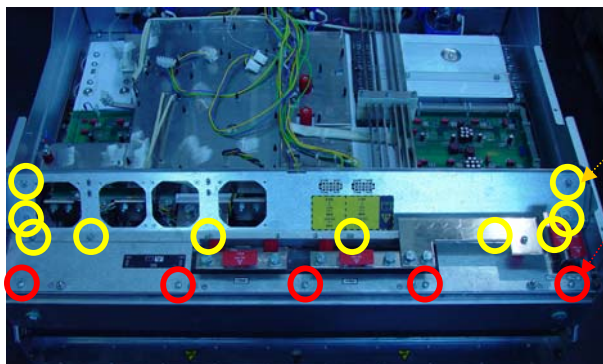


Step 2

Remove 4 screws (P1).



Push on the 2 plastic supports
to remove the connector X2.



Step 3

Remove 10 screws (P2).

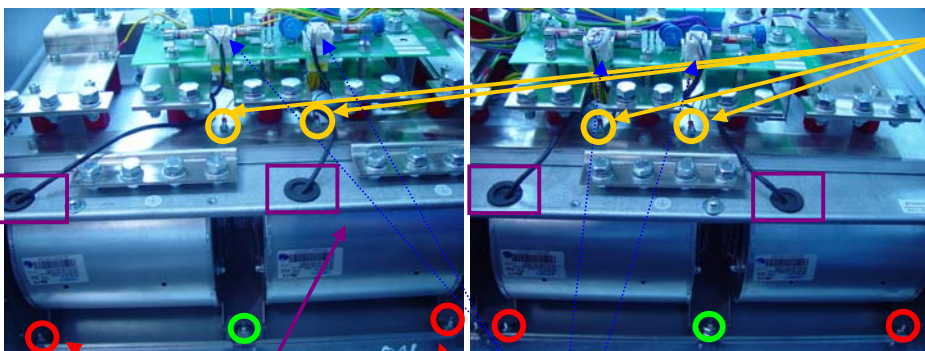
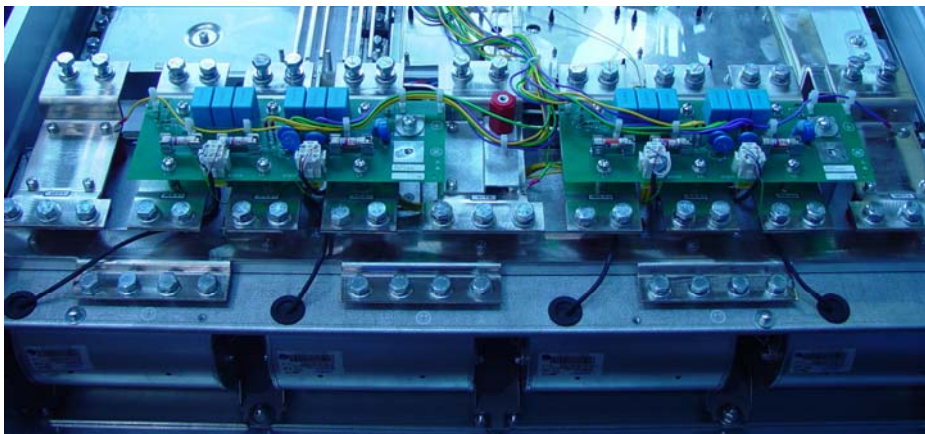
Remove 5 screws (P1)



Mark	Size	Torque
P3	M12x25	45Nm
P1	M6x14	5.5Nm
P2	M6	5.5Nm

The Metal Body Parts KIT can be removed.

Fan Power Electronic. VZ3V1212



Be careful; don't forget to disconnect the ground wire.

Disconnect 4 connectors

Fan turbine 1->X11 on RFI filter board 1->X33 on Braking unit device

Fan turbine 2->X12 on RFI filter board 1->X33 on Braking unit device

Fan turbine 3->X13 on RFI filter board 2->X3 on fan control board 1

Fan turbine 4->X14 on RFI filter board 2->X3 on fan control board 1

Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

Remove 4 screws (J130)

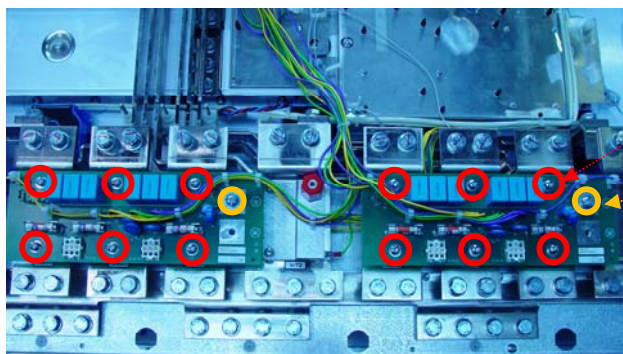
Remove 2 screws (J139)

The Turbine can be removed.



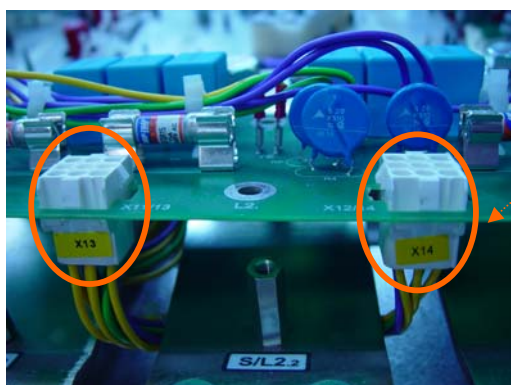
Mark	Size	Torque
J130	M6	5.5Nm
J139	M8	13.5Nm

RFI Filter Board. VX4A1121



Remove 12 screws (P1)

Remove 2 screws (P2)



To change the board, don't forget to disconnect the 4 connectors

And
X11 X12 on RFI Filter Board 1

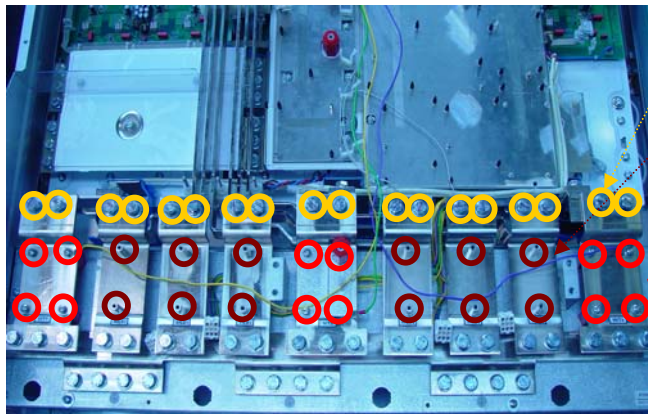
X13 X14 on RFI Filter Board 2

The Both RFI filter board can be removed.



Mark	Size	Torque
P2	M6x12	5.5Nm
P1	M6x14	5.5Nm

AC Bus Bar Kit: VZ3N1346



Remove 18 screws (P1)

Remove 12 screws (P2)

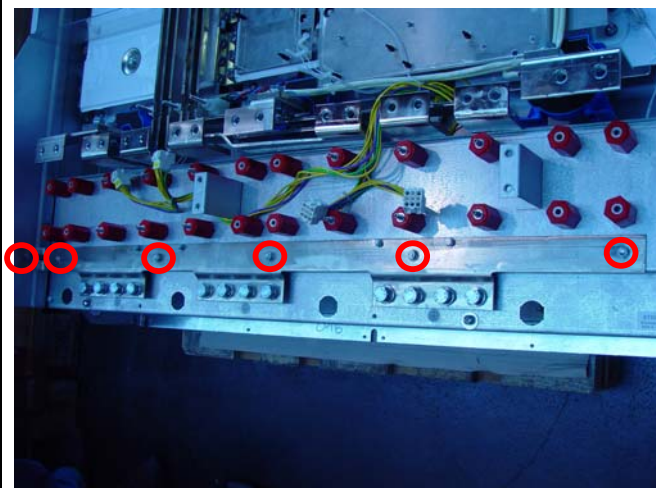
Remove 12 screws (P3)



Mark	Size	Torque
P1	M12x25	45Nm
P2	M6x22	5.5Nm
P3	M6x14	5.5Nm

The AC Bus Bar can be removed.

Earth Terminal Bar: VZ3N1346



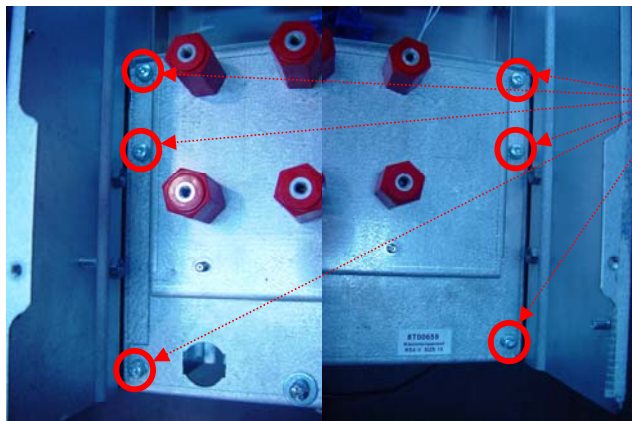
Remove 6 nuts



Size	Torque
M6	5.5Nm

The Ground bar can be removed.

Metal Body Part KIT



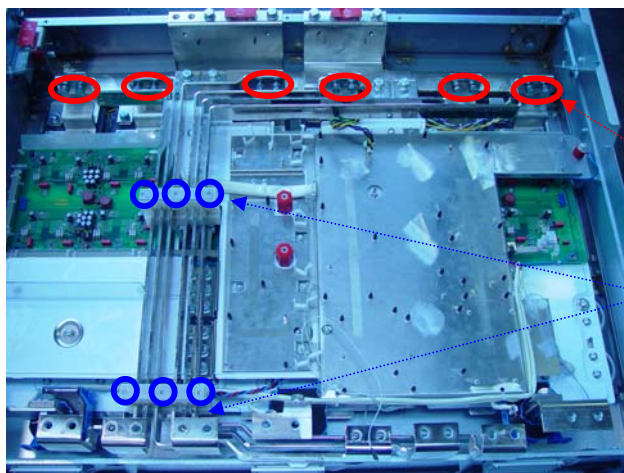
Remove 6 screws



Size	Torque
M6x14	5.5Nm

The Metal body parts KIT can be removed.

AC Bus Bar Kit: VZ3N1346



Remove 12 screws.

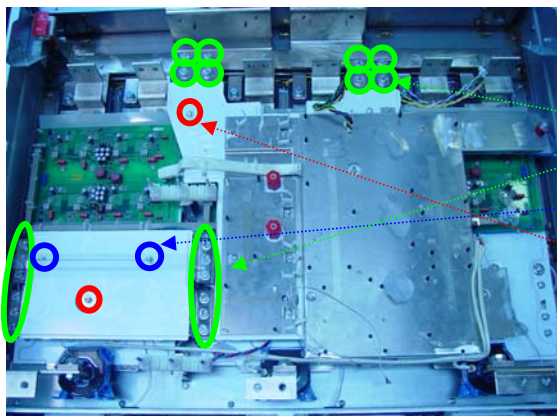
Remove 6 nuts



Size	Torque
M12x25	45Nm
M6	5.5Nm

The AC Bus Bars kit can be removed.

DC bus bar kit: VZ3N1343

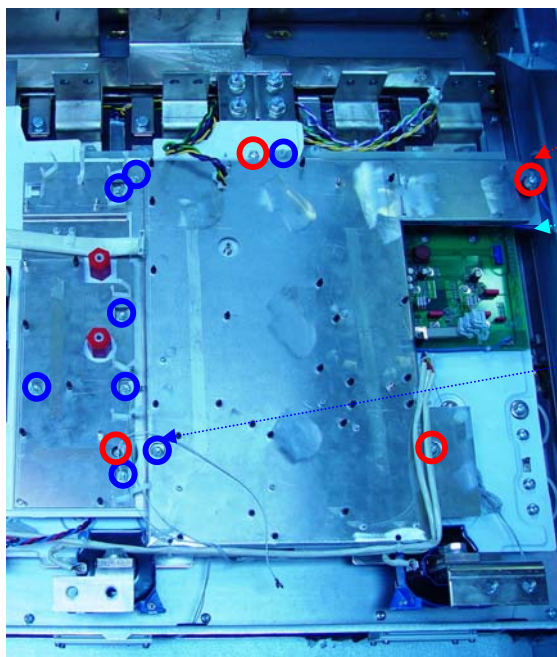


Step 1

Remove 20 screws (J129).

Remove 2 screws (P2)

Remove 2 screws (P2)



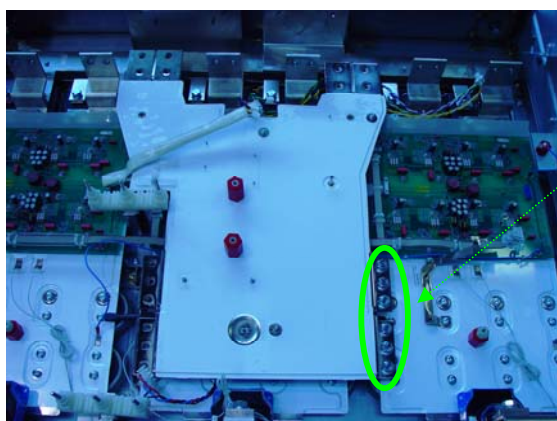
Step 2

Remove 4 screws (P2)

And disconnect one wire:

Blue -> X53 on gate drive board IGBT W

Remove 8 nuts (P3)



Step 3

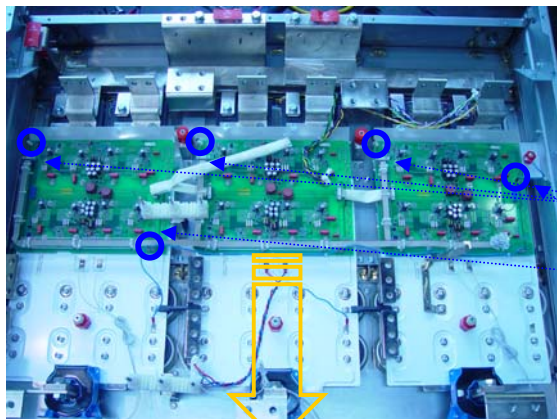
Remove 6 screws (J129)



Mark	Size	Torque
J129	M10x20	27Nm
P2	M6X12	5.5Nm
P3	M6	5.5Nm

The DC bus bar kit can be removed.

Gate Drive Board IGBT.VX5A1207

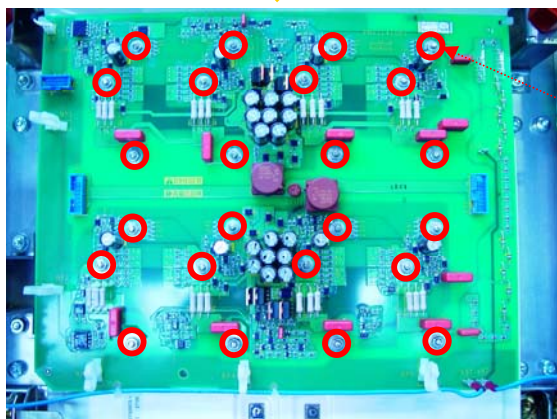


Disconnect the 4 wires top to down.

Black X4 on gate drive Board IGBT U/V/W-> thermal sensor

Blue X53 on gate drive board IGBT W-> DC bus

Blue X51 on gate drive Board IGBT U->Power Interconnection bar U



For each 3 branch U/V/W, you should remove:

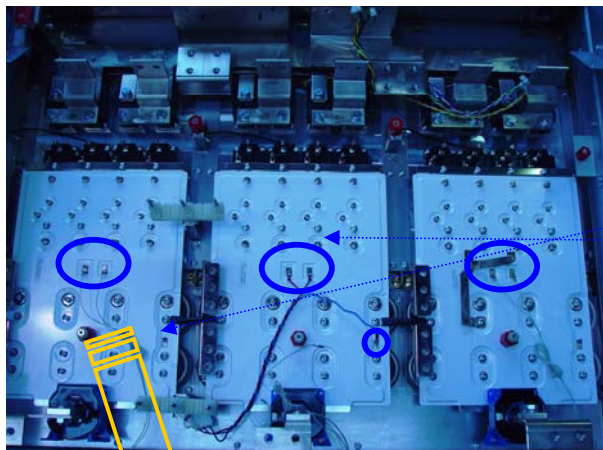
Remove 24 nuts

The Gate Drive Boards IGBT can be removed.



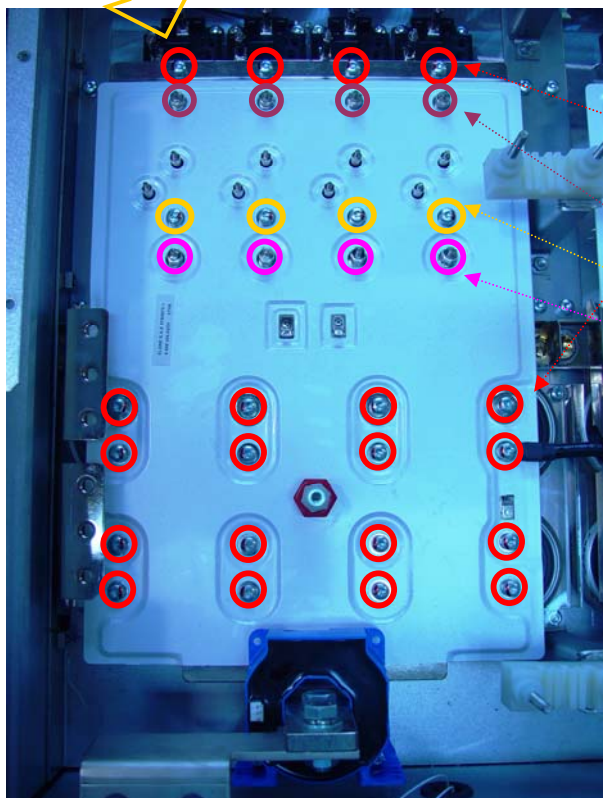
Size	Torque
M4	1.2Nm

DC bus bar kit: VZ3N1343



Disconnect the 4 wires Left to right.

White wires on interconnection bar U->discharging resistor
Red/Black/Blue wires on interconnection bar V->X1 on active current balancing Board
White wires on interconnection bar W->discharging resistor



For 3 AC bus bar U V, you should remove:

Remove 4 screws (P1)

Remove 16 screws (P2-P3-P4).

Remove 4 screws (P6)

Remove 4 screws (P7)

Remove 4 screws (P5)

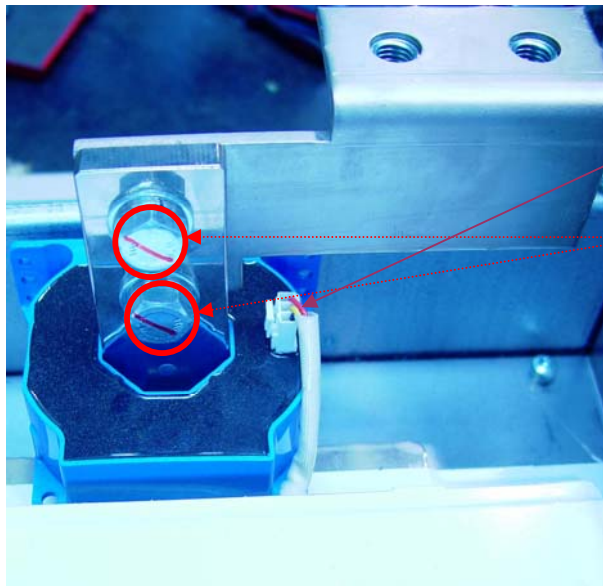
Be careful, at the time of reassembling;
don't forget to put the 2 Shunt.

The DC Bus Bar-Phase can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2-P3-P4	M6x14	3,3Nm
P6	ST M6/M4	5.5Nm
P7	M6x20	5.5Nm
P5	ST M6/M4	5.5Nm

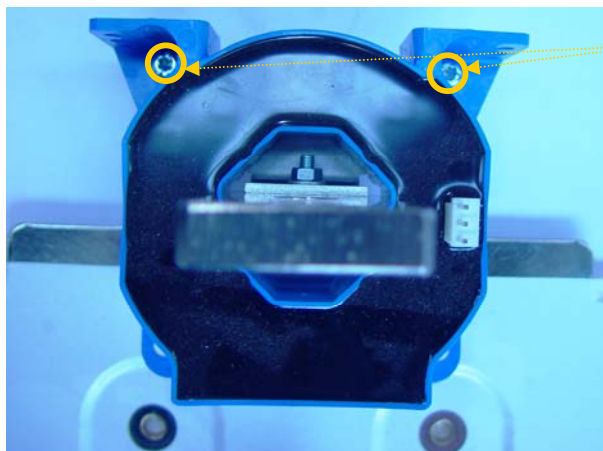
Current sensor: VY1A1109



For 3 current sensor on U/V/W, you should remove:

Disconnect one wire:
Current sensor -> X11 on measuring board

Remove 2 screws (P1)



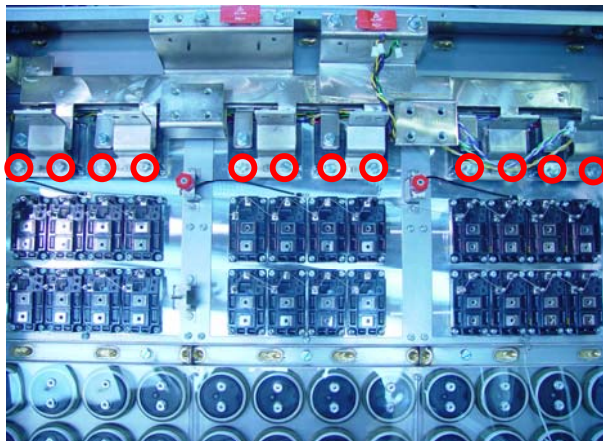
Remove 2 screws (P2)

The current sensor can be removed.



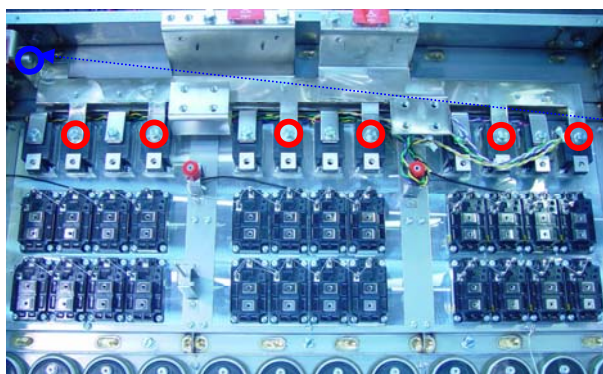
Mark	Size	Torque
P1	M12x25	45Nm
P2	M4x12	1.2Nm

Rectifier module thyristor: VZ3TM2200M4371
Rectifier module diode: VZ3DM2200M5471
AC Bus Bar Kit: VZ3N1346



Part 1

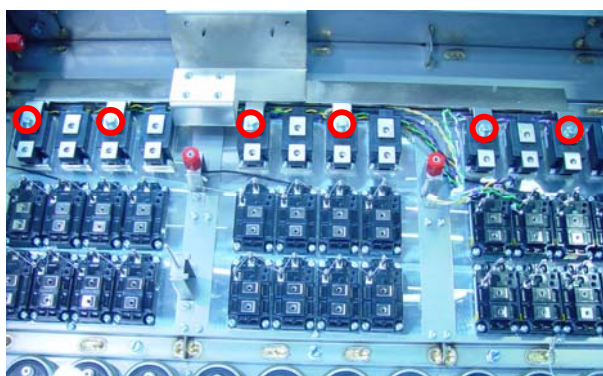
Remove 12 screws (P2)



Part 2

Remove 6 screws (P2)

Remove 1 screw (P1)



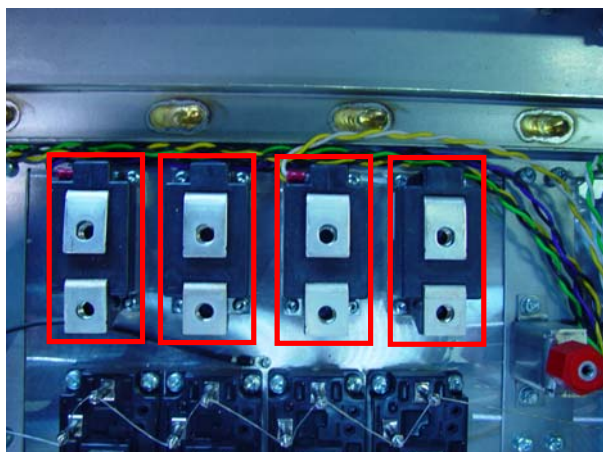
Part 3

Remove 6 screws (P2)



Mark	Size	Torque
P2	M10x25	13.5Nm
P1	M6x12	5.5Nm

Rectifier module thyristor: VZ3TM2200M4371
Rectifier module diode: VZ3DM2200M5471



For each 3 branch, you should remove:

Remove 16 screws

It is **not necessary** to change all rectifiers but just this one that is damaged.

After changing, **be careful** to set up direction.
Don't forget applying the grease.



Size	Torque
M5x25	5.0Nm

Rectifier module thyristor: VZ3TM2200M4371



Be careful, after changing respect the colour line for the wires.

From left to right

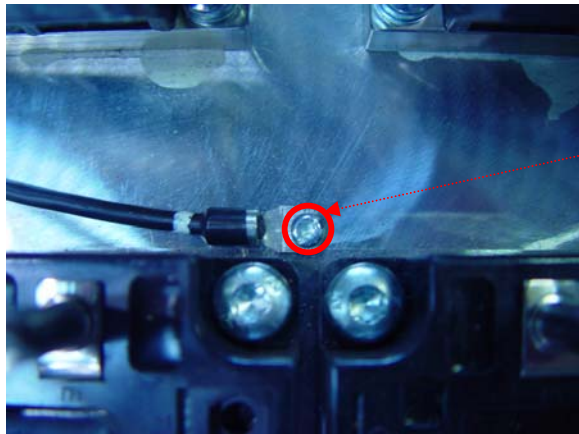
[**Yellow**-Black]- [**Green**-Black] - [**Purple**-Black]

And

[**Yellow**-White]- [**Green**- White] - [**Purple**- White]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Thermal Sensor. VZ3G1105



For each 3 branch U/V/W, you should remove:

Remove 1 screw.

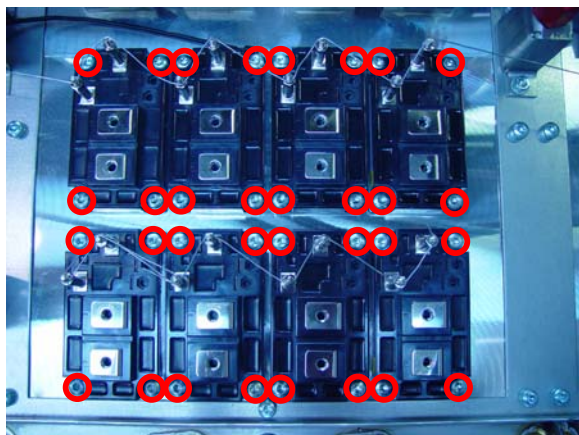
Be careful, don't forget to connect the wire on the connector "X4" of the Gate drive board IGBT



Size	Torque
M3x6	0.8Nm

The thermal sensor can be removed.

IGBT modules: VZ3IM1504M1771



For each 3 branch U/V/W, you should remove:

Remove 32 screws.

It is **not necessary** to change all IGBT but just this one that is damaged.

After changing, **be careful** to set up direction.

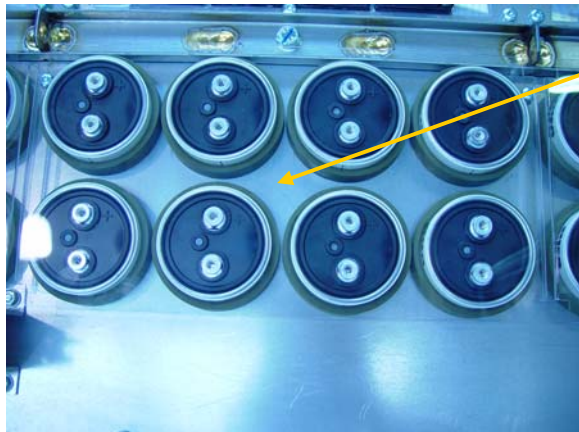
Don't forget applying the grease.



Size	Torque
M6x20	3.0Nm

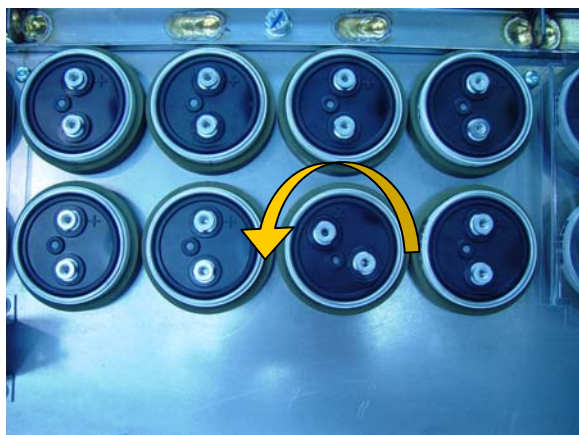
The IGBT can be removed.

Lot of Capacitors. VY1ADC1117



First step

Remove plastic protection part **for each 3 branch** U/V/W



Second step

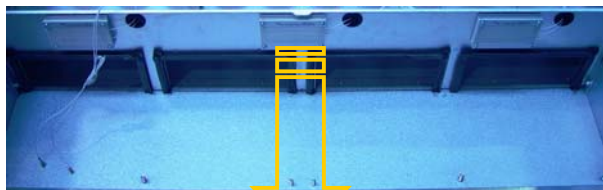
For each 3 branch U/V/W, you should remove 6 capacitors:

To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction

The Capacitors can be removed.

Discharging Resistor: VZ3R36KW125



For each 3 discharging resistor, you should remove:

Remove 4 screws.



Be careful, to remove the discharging resistor, remove the gasket and slide the wire in the hole.

The discharging resistor can be removed.



Size	Torque
M4x20	1.2Nm



9.30.2 Product Assembling Drawing

No information

9.30.3 Product Cabling Drawing

Refer to following file: [Cabling diagram size 15.pdf part1](#)

Refer to following file: [Cabling diagram size 15.pdf part2](#)

10. Procedure for validation after repair

10.1 *ATV71 sizes 2 to 15*

Once the product has been reassembled, connect the control test circuit, and then switch on the product:

- Run the **diagnostic procedure**
- If it is OK, run to the test bench with **short acceleration** and **short deceleration**
- If it is OK, transfer the new PC configuration to the drive
- Use the setup software (list of parameters) to edit it

10.2 Quality reporting to the **VSD division**

Information must be entered in the computer system set up in each country-based organisation so that the **VSD division** can manage product quality as well as possible. The repair report should include a description of the malfunction, the diagnostics and the fault discovered.

11. Structure Drives

11.1 Common Part: Control Block

It's compound of 4 boards, and it is common for all ratings.

- Display board
- Terminal board
- Application board
- Motor Control board

11.2 Electronic & mechanical structure of all drives

Power rating :	ATV71H037M3X, ATV71H075M3X, ATV71HU15M3X, ATV71H075N4, ATV71HU15N4, ATV71HU22N4 ATV61H075M3X, ATV61HU15M3X, ATV61H075N4, ATV61HU15N4, ATV61HU22N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Current measurement (shunts)
	Fan driver
	Pre-charge relay driver
	EEPROM
	Motor output terminal
Bus board :	Function:
	Main capacitors
	Pre-charge resistor
Filter board :	Functions :
	Input power terminal
	EMC Filter
Power module	
Fan	
ZCT: Ground fault detection	

Power rating :	ATV71HU22M3X, ATV71HU30M3X, ATV71HU40M3X, ATV71HU30N4, ATV71HU40N4 ATV61HU22M3X, ATV61HU30M3X, ATV61HU40M3X, ATV61HU30N4, ATV61HU40N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Current measurement
	Fan driver
	Pre-charge relay driver
	EEPROM
	Motor output terminal
Bus board :	Function:
	Main capacitors
	Pre-charge resistor
Filter board :	Functions :
	Input power terminal
	EMC Filter
Power module	
Fan	
ZCT: Ground fault detection	

Power rating :	ATV71HU55M3X, ATV71HU55N4, ATV71HU75N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Current measurement
	Fan driver
	Pre-charge relay driver
Bus board :	EEPROM
	Function:
Filter board :	Main capacitors
	Functions :
	Input power terminal
EMC Filter	
Power module	
Pre-charge relay and resistor.	
Motor output terminal.	
Fan	
ZCT: Ground fault detection	
Mechanical structure: plastic	

Power rating :	ATV71HU75M3X, ATV71HD11N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Current measurement
	Fan driver
	Pre-charge relay driver
Bus board :	EEPROM
	Function:
Filter board :	Main capacitors
	Functions :
	Power filtrate
EMC Filter	
Power module	
Pre-charge relay and resistor.	
Motor output terminal.	
Fan	
ZCT: Ground fault detection	
Mechanical structure: plastic	

Power rating :	ATV71HD11M3X, ATV71HD15M3X, ATV71HD15N4, ATV71HD18N4 ATV61HD11M3X, ATV61HD15M3X, ATV61HD15N4,ATV61HD18N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Current measurement
	Fan driver
	Pre-charge relay driver
Bus board :	Motor output terminal
	EEPROM
Filter board :	Function:
	Pre-charge relay.
	Pre-charge resistor.
Main capacitors	
Filter board :	Functions :
	Input power terminal
	EMC Filter
Power module	
Fan	
ZCT: Ground fault detection	
Mechanical structure: plastic	

Power rating :	ATV71HD18M3X, ATV71HD22M3X, ATV71HD22N4 ATV61HD18M3, ATV61HD22M3, ATV61HD22N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Current measurement
	Fan driver
	Pre-charge relay driver
Bus board :	Motor output terminal
	EEPROM
Filter board :	Function:
	Pre-charge relay (only 3-5.5KW).
	Pre-charge resistor (only 3-5.5KW).
Main capacitors	
Filter board :	Functions :
	Input power terminal
	EMC Filter
Power module	
Fan	
ZCT: Ground fault detection	
Mechanical structure: plastic	
Pre-charge relay (only 7.5kW).	
Pre-charge resistor (only 7.5kW).	

Power rating :		ATV71HD30N4, ATV71HD37N4, ATV61HD30N4, ATV61HD37N4 ATV71HD30M3X, ATV71HD37M3X, ATV71HD45M3X, ATV61HD30M3X, ATV61HD37M3X, ATV61HD45M3X
HMI board		
Power board :	Functions:	
	Motor control : Function	
	Supply voltage	
	Gate driver	
	Fan driver	
	Pre-charge relay driver	
Bus board :	EEPROM	
	Function:	
	Pre-charge resistor	
Filter board :	Power module	
	Functions :	
	EMC Filter	
Interconnect board: between power board and Bus board.		
Main capacitors		
Pre-charge relay		
Fan		
Current measurement		
Motor output terminal		
Mechanical structure: sheet metal.		

Power rating :		ATV71HD30M3X, ATV71HD37M3X, ATV71HD45M3X
HMI board		
Power board :	Functions:	
	Motor control : Function	
	Supply voltage	
	Gate driver	
	Fan driver	
	Pre-charge relay driver	
Bus board :	EEPROM	
	Function:	
Resistor board:	Function	
	Pre-charge resistor	
Filter board :	Relay	
	Functions :	
	EMC Filter	
Interconnect board: between power board and Bus board.		
Main capacitors		
Power module		
Pre-charge relay		
Fan		
Current measurement		
Motor output terminal		
Mechanical structure: sheet metal.		

Power rating :	ATV71HD45N4, ATV71HD55N4, ATV71HD75N4 ATV61HD45N4, ATV61HD55N4, ATV61HD75N4
HMI board	
Power board :	Functions: Motor control : Function Supply voltage Gate driver Fan driver Pre-charge relay driver EEPROM
Bus board	
Resistor board:	Function Pre-charge resistor Relay
Filter board :	Functions : EMC Filter
Snubber board :	Function
Interconnect board: between power board and Bus board.	
Main capacitors	
Power module	
Pre-charge relay	
Fan	
Current measurement	
Motor output terminal	

Power rating :	ATV71HD55M3X, ATV71HD90N4 ATV61HD55M3X, ATV61HD75M3X, ATV61HD90N4, ATV61HC11N4
HMI board	
Power board :	Functions: Motor control : Function Supply voltage Gate driver Fan driver Pre-charge relay driver EEPROM
Bus board :	Function: Pre-charge resistor Power module
Filter board 1 :	Functions : EMC Class-A Filter
Filter board 2 :	Functions : EMC Class-A Filter
Filter board 3 :	Functions : EMC Class-B Filter
Filter	Functions : EMC Class-B Filter
Main capacitors	
Pre-charge relay	
Fan	
Current measurement	
Motor output terminal	
Mechanical structure: sheet metal.	

Power rating :	ATV71HD75M3X, ATV71HC11N4, ATV61HD90M3, ATV61HC13N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Pre-charge relay driver
	EEPROM
Bus board	
Resistor board :	Function
	Pre-charge resistor
Filter board 1 :	Functions :
	EMC Class-A Filter
Filter board 2 :	Functions :
	EMC Class-B Filter
Filter	Functions :
	EMC Class-B Filter
PSU board :	Function
Fan driver	
Power module	
Main capacitors	
Pre-charge relay	
Current measurement	
Motor output terminal	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC13N4, ATV61HC16N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Pre-charge relay driver
	EEPROM
Bus board	
Resistor board :	Function
	Pre-charge resistor
	Relay
Filter board 1 :	Functions :
	EMC Class-A Filter
Filter board 2 :	Functions :
	EMC Class-B Filter
Filter board 3 :	Functions :
	EMC Class-B Filter
Filter	Functions :
	EMC Class-B Filter
PSU board :	Function
Fan driver	
Power module	
Main capacitors	
Pre-charge relay	
Current measurement	
Motor output terminal	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC16N4, ATV61HC22N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function
	Main capacitors
PSU board	Function :
	Fan driver
Filter	Functions :
	EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC20N4, ATV71HC25N4, ATV71HC28N4, ATV61HC25N4, ATV61HC31N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function
	Main capacitors
PSU board	Function :
	Fan driver
Filter	Functions :
	EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC31N4, ATV71HC40N4, ATV61HC40N4, ATV61HC50N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC50N4, ATV61HC63N4
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV61WD30N4, ATV61WD30N4C, ATV71WD22N4, ATV71WD22N4C
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV61WD37N4, ATV61WD45N4, ATV61WD37N4C, ATV61WD45N4C ATV71WD30N4, ATV71WD37N4, ATV71WD30N4C, ATV71WD37N4C
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV61WD35N4, ATV61WD75N4, ATV61WD90N4, ATV61WD55N4C, ATV61WD75N4C, ATV61WD90N4C, ATV71WD45N4, ATV71WD55N4, ATV71WD75N4, ATV71WD45N4C
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HD22Y, ATV71HD30Y, ATV71HU40Y, ATV71HU55Y, ATV71HU75Y, ATV71HD11Y, ATV71HD15Y, ATV71HD18Y, ATV71HD22Y, ATV71HD30Y, ATV61HU30Y, ATV61HU40Y
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HD37Y, ATV71HD45Y, ATV71HD55Y, ATV71HD75Y, ATV71HD90Y, ATV61HD37Y, ATV61HD45Y, ATV61HD55Y, ATV61HD75Y, ATV61HD90Y
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC11Y, ATV71HC13Y, ATV71HC16Y, ATV61HC11Y, ATV61HC13Y, ATV61HC16Y, ATV61HC22Y
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC20Y, ATV71HC25Y, ATV71HC31Y, ATV61HC25Y, ATV61HC31Y, ATV61HC40Y
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

Power rating :	ATV71HC40Y, ATV71HC50Y, ATV71HC63Y, ATV61HC50Y, ATV61HC63Y, ATV61HC80Y
HMI board	
Power board :	Functions:
	Motor control : Function
	Supply voltage
	Gate driver
	Fan driver
	Current measurement
	Pre-charge relay driver
	Pre-charge relay
	Pre-charge resistor
	EMC filter
	ZCT:Ground fault detection
	Motor output terminal
	EEPROM
Filter board:	Function Main capacitors
PSU board	Function : Fan driver
Filter	Functions : EMC Class-B Filter
Power module	
Mechanical structure: sheet metal.	

12. Inspection and Maintenance

	Danger
Mandatory	- The equipment must be inspected every day. If the equipment is not inspected and maintained, errors and malfunctions may not be discovered which could lead to accidents. - Before inspection, perform the following steps. (1) Shut off all input power to the inverter. (2) Wait at least ten minutes and check to make sure that the charge lamp is no longer lit. (3) Use a tester that can measure DC voltages (800V DC or more), and check that the voltage to the DC main circuits (across PA-PC) does not exceed 45V. Performing an inspection without carrying out these steps first could lead to electric shock.

Be sure to inspect the inverter regularly and periodically to prevent it from breaking down because of the environment of use, such as temperature, humidity, dust and vibration, or deterioration of its components with aging.

12.1 Regular inspection

Since electronic parts are susceptible to heat, install the inverter in a cool, well-ventilated and dust-free place. This is essential for increasing the service life. The purpose of regular inspections is to maintain the correct environment of use and to find any sign of failure or malfunction by comparing current operation data with past operation records.

Subject of inspection	Inspection procedure			Criteria for judgement
	Inspection item	Inspection cycle	Inspection method	
1. Indoor environment	1) Dust, temperature and gas 2) Drop of water or other liquid 3) Room temperature	Occasionally Occasionally Occasionally	1) Visual check, check by means of a thermometer, smell check 2) Visual check 3) Check by means of a thermometer	1) Improve the environment if it is found to be unfavorable. 2) Check for any trace of water condensation. 3) Max. temperature: 60°C
2. Units and components	1) Vibration and noise	Occasionally	Tactile check of the cabinet	Is something unusual is found, open the door and check the transformer, reactors, contactors, relays, cooling fan, etc., inside. If necessary, stop the operation.
3. Operation data (output side)	1) Load current 2) Voltage (*) 3) Temperature	Occasionally Occasionally Occasionally	Moving-iron type AC ammeter Rectifier type AC voltmeter Thermometer	To be within the rated current, voltage and temperature. No significant difference from data collected in a normal state.

*) The voltage measured may slightly vary from voltmeter to voltmeter. When measuring the voltage, always take readings from the same circuit tester or voltmeter.

Check points

1. Something unusual in the installation environment.
2. Something unusual in the cooling system.
3. Unusual vibration or noise.
4. Overheating or discoloration.
5. Unusual.
6. Unusual motor vibration, noise or overheating.
7. Adhesion or accumulation of foreign substances (conductive substances).

Cautions about cleaning

To clean the inverter, wipe dirt off only its surface with a soft cloth but do not try to remove dirt or stains from any other part. If stubborn stains persist, remove them by wiping gently with a cloth dampened with neutral detergent or ethanol. Never use any of the chemicals in the table below; the use of any of them may damage or peel the coating away from moulded parts (such as plastic covers and units) of the inverter.

Acetone	Ethylene chloride	Tetrachloroethane
Benzen	Ethyl acetate	Trichloroethylene
Chloroform	Glycerin	Xylene

12.2 Periodical inspection

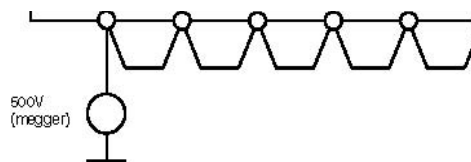
Make a periodical inspection at intervals of 3 or 6 months depending on the operating conditions.

Danger	
Mandatory	• Before inspection, perform the following steps. (1) Shut off all input power to the inverter. (2) Wait at least ten minutes and check to make sure that the charge lamp is no longer lit. (3) Use a tester that can measure DC voltages (800V DC or more), and check that the voltage to the DC main circuits (across PA-PC) does not exceed 45V. Performing an inspection without carrying out these steps first could lead to electric shock.
Prohibited	• Never replace any part. This could be a cause of electric shock, fire and bodily injury. To replace parts, call the local sales agency.

Check items

1. Check to see if all screwed terminals are tightened firmly. If any screw is found loose, tighten it again with a screwdriver.
2. Check to see if all caulked terminals are fixed properly. Check them visually to see that there is no trace of overheating around any of them.
3. Check all cables and wires for damage. Check them visually.
4. Remove dirt and dust. With a vacuum cleaner, remove dirt and dust. When cleaning, clean the vents and the printed circuit boards. Always keep them clean to prevent an accident due to dirt or dust.
5. If no power is supplied to the inverter for a long time, the performance of its capacity electrolytic capacitor declines. When leaving the inverter unused for a long time, supply it with electricity once every two years, for 5 hours or more each, to recover the performance of the capacity electrolytic capacitor. And also check the function of the inverter. It is advisable not to supply the commercial power directly to the inverter but to gradually increase the power supply voltage with a transformer, etc.
6. If the need arises, conduct an insulation test on the main circuit terminal board only, using a 500V insulation tester. Never conduct an insulation test on control terminals other than terminals on the printed circuit board or on control terminals. When testing the motor for insulation performance, separate it from the inverter in advance by disconnecting the cables from the inverter output terminals U, V and W. When conducting an insulation test on peripheral circuits other than the motor circuit, disconnect all cables from the inverter so that no voltage is applied to the inverter during the test.

(Note) Before an insulation test, always disconnect all cables from the main circuit terminal board and test the inverter separately from other equipment.



7. Never test the inverter for pressure. A pressure test may cause damage to its components.
8. Voltage and temperature check
Recommended voltmeter: Input side ... Moving-iron type voltmeter

Output side ... Rectifier type voltmeter



It will be very helpful for detecting a defect if you always measure and record the ambient before, during and after the operation.

Replacement of expendable parts

The inverter is composed of a large number of electronic parts including semiconductor devices. The following parts deteriorate with the passage of time because of their composition or physical properties. The use of aged or deteriorated parts leads to degradation in the performance or a breakdown of the inverter. To avoid such trouble, the inverter should be checked periodically.

Note) generally, the life of a part depends on the ambient temperature and the conditions of use. The life spans listed below are applicable to parts when used under normal environmental conditions.

1) Cooling fan the fan, which cools down heat-generating parts, has a service life of about 30,000 hours (about 2 or 3 years of continuous operation). The fan also needs to be replaced if it makes a noise or vibrates abnormally.

2) Smoothing capacitor

- Main circuit.
The smoothing capacitors applied to this inverter in the main circuit DC section are film type capacitors. Those lifetimes in design, value is 15 years, but it is recommended to replace them after it is used for about 10 years under normal conditions. Since the smoothing capacitors are mounted on a printed circuit board, it needs to be replaced together with the circuit board.
- Control circuit.
The smoothing aluminium electrolytic capacitor degrades in performance because of ripple current, etc. The lifetime in design value is 15 years, but it is recommended to replace them after it is used for about 10 years under normal conditions. Since the smoothing capacitors are mounted on a printed circuit board, it needs to be replaced together with the circuit board.
<Criteria for appearance check>
 - Absence of liquid leak
 - Absence of case damp aged

Note: The operation time is helpful for roughly determining the time of replacement. For the replacement of parts, contact your nearest Schneider Electric distributor. For safety's sake, never replace any part on your own. (Parts replacement alarms can be known by monitor and alarm output, if it is set. Refer to section 6.17.14).

Standard replacement cycles of principal parts

As guides, the table below lists part replacement cycles that were estimated based on the assumption that the inverter would be used in a normal use environment under normal conditions (ambient temperature, ventilation conditions, and energizing time). The replacement cycle of each part does not mean its service life but the number of years over which its failure rate does not increase significantly.

Part name	Standard replacement cycle	Replacement mode and others
Cooling fan	2 to 3 years	Replacement with a new one
Main circuit smoothing electrolytic capacitor	10 years	Replacement with a new one
Relay and contactor	-	Whether to replace or not depends on the check results
Aluminum electrolytic capacitor mounted on a printed circuit board	10 years	Replace with a new circuit board

Note) the life of a part greatly varies depending on the environment of use.

12.3 Keeping the drive in the storage

Take the following precautions when keeping the inverter in storage temporarily or for a long period of time.

- 1 Store the inverter in a well-ventilated place away from heat, damp, dust and metal powder.
- 2 If the printed circuit board in your inverter has an anti-static cover (black cover), do not leave it detached from the circuit board during storage. The cover must be detached before turning on the inverter.
- 3 If no power is supplied to the inverter for a long time, the performance of its capacity electrolytic capacitor declines. When leaving the inverter unused for a long time, supply it with electricity once every two years, for 5 hours or more each, to recover the performance of the capacity electrolytic capacitor. And also check the function of the inverter. It is advisable not to supply the commercial power directly to the inverter but to gradually increase the power supply voltage with a transformer, etc.

12.4 Disposal of the drive

	Danger
Mandatory	• If you throw away the inverter, have it done by a specialist in industry waste disposal(*). If you throw away the inverter by yourself, this can result in explosion of capacitor or produce noxious gases, resulting in injury. (*) Persons who specialize in the processing of waste and known as "industrial waste product collectors and transporters" or "industrial waste disposal persons. "If the collection, transport and disposal of industrial waste is done by someone who is not licensed for that job, it is a punishable violation of the law. (Laws in regard to cleaning and processing of waste materials)

For safety's sake, do not dispose of the disused inverter yourself but ask an industrial waste disposal agent. Disposing of the inverter improperly could cause its capacitor to explode and emit toxic gas, causing injury to persons.
